AWI-ESM3 ↔ PISM moving-cavity coupling

the investigation report

Single living document. Superseded claims are struck through, not deleted.

Last updated: 2026-07-31

Abstract

Mission. A working end-to-end AWI-ESM3 (FESOM2–CORE3-cavity + OIFS-48r1 + LPJ-GUESS + WAM) ↔ Antarctic-PISM cycle: awiesm3 year → PISM decade → awiesm3 year on a changed mesh, with all three coastline-following members (FESOM cavity, OIFS land–sea mask, WAM sea points) regenerated per leg.

Where we are. Eight real bugs and one architectural redesign (the coupled slab) later, the cycle is proven end-to-end with real cavities — awiesm3 year → PISM decade → awiesm3 year on the changed mesh, plus one further clean cycle. The 10-year production attempt then ran eleven more increments *with the cavities accidentally amputated by an overreaching mesh-hygiene fix* (discovered at wrap-up, reverted; those legs are tainted). The campaign continues, and the diagnosis has turned to physics: the Ross-front blowup is increment magnitude, not coastline fragility, and the increments are large because the PI-control cavity melt runs ∼ 7× observations. That melt is traced to cavity currents ∼ 6× too fast — and a standalone dissection (§1.13) then shows those fast currents are *injected by the mesh increment*: the geometry and the water are both fine (a cold start on the changed mesh settles to the healthy ∼2.6 cm/s), but the *remapped restart is internally inconsistent with the new mesh* through `hnode` (the ALE layer thickness): the remap's ALE-rescale step compresses the changed-column `hnode` using a stale `eta`, seeding a barotropic spin-up. Fixing that (nominal `hnode` at the changed columns) is a small `remap_hnode` edit and *halves* the over-speed (17→10 cm/s in isolation), but does not fully restore health: reaching ∼4 cm/s needs nominal `hnode` over the *whole* ocean, which points not at the cavity fill but at the **vertical-coordinate seam** — zstar (stretched layers) outside vs. the rigid-lid cavity (nominal layers) inside. `linfs` (fixed layers everywhere) removes it: the raw remap under `linfs` does not spin up (decays to ∼1.7 cm/s, no `hnode` surgery) — the clean complete fix, if a linearised free surface is acceptable model-wide (§1.13, §1.14). This is confirmed in the coupled chain, and then closed: with `linfs` plus a fill fix for a separate artifact (newly-iced columns inheriting warm open-ocean water, cured by seeding open→cavity columns from the nearest existing cavity), the `movcav37` run carries the full decade 1900–1909 — **ten coupled years, nine mesh increments, cavity current held at 1–2 cm/s throughout, no spin-up and no blowup**, cavity preserved, melt in the observed range (§1.15, §1.16). Rolling the coupled slab out to the fixed-mesh siblings then caught a months-old silent bug — `develop/develop-cc` never set the OIFS `NAMMCC` switches, so the coupled ice thickness was discarded and their Arctic went ice-free every summer; fixed in config, the 10-year sea-ice climate validates against an independent CMIP7 spin-up (§1.18). The mesh-increment crash family (`voskin:282`, Antarctic) is now inventoried and instrumented, reproduction pending (§1.19). Of the two further crash faces that instrumentation exposed, the leg-start one is **solved**: not the SSH solve it looked like, but **Bug VIII** — `fer_solve_Gamma` publishing an unassigned automatic-array element into the GM streamfunction wherever a node's surrounding elements invert its solve range, giving a 10^{251} bolus velocity that detonates the tracer solve. A cavity front on steep bathymetry is what creates that node geometry, which is why a decade of FESOM never hit it (§1.21).

Contents

1 The campaign, in the order it happened

Each subsection carries the date it was worked. The bug numbering (I–V) is the order of *discovery* and is kept as the canonical naming.

1.1 The restart seam and the cold-start control (2026-07-08–13)

The original FESOM cavity-margin blowup: a remapped warm restart mixes an evolved, balanced dynamic state (98% of nodes, bit-exact) with patched values at changed nodes; the *seam* between them is the instability. The decisive control:

Run	Δt	Initial state	Outcome
full-mesh control	1200 s	native spun-up restart	stable, worst CFLz 2.61
PISM cavity, cold	**1200 s**	**cold start**	**stable, CFLz 3.44**
PISM cavity, warm	1200 s	remapped restart	dies day 4.0 (CFLz 9.2)
PISM cavity, warm	60 s	remapped restart	dies day 4.4 (η_n, 0 CFLz)

The ocean lives happily in PISM's cavity; the production answer became: run chunk-1 cold-started on the PISM cavity (artifacts pooled), and let `remap_restart` carry only genuine per-leg increments. The first real increment test crashed on 17 fully-opened columns (ice removed entirely, 16 levels at once) out of 1399 — but its numbers are TAINTED (measured under Bugs I+II) and the finding that matters survived into the fixes below: absorbing 99% of a geometry change is not a passing grade when the remaining 1% is fatal.

1.2 Bug I — PISM was fed surface water as ice-base forcing (2026-07-14)

`coupling_ocean2pism` built PISM's `-ocean th` forcing as a mean over FESOM level *indices* 1–20 = **0–410 m including the summer surface** (the correct `-sellevel,150/600` sat in dead code). PISM received ice-base water up to +6.5°C even from a healthy ocean, melted at ∼30 m/yr, and shed **24% of its floating area per decade** — every mesh-change leg then had to absorb a catastrophic geometry increment. FESOM's own cavity melt in the same runs was sane (2.2 m/yr mean, 0.72 median).

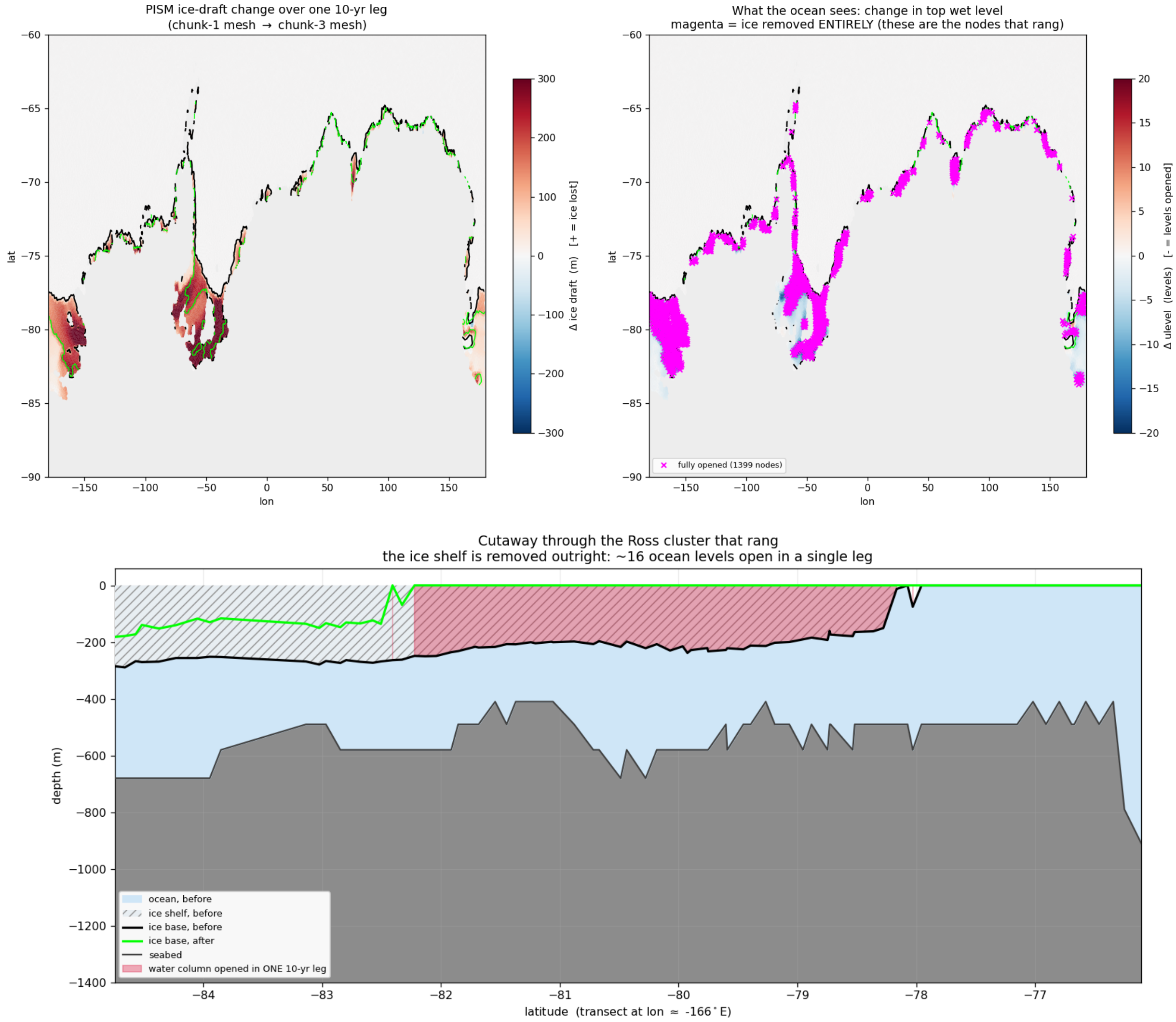

Figure 1: What Bug I did to PISM in one 10-year leg under surface-water forcing: 1399 nodes lose their ice entirely (magenta ring around the whole margin), the Ross front retreats ≈440 km (section: ice base $-200\ldots-280$ m → open ocean across four degrees of latitude).

Fix (`93a207ee`): the DIRECT route. PISM consumes FESOM's own cavity fluxes via `-ocean given`: `fw`→`shelfbmassflux` (×1000, sign preserved), `sst`→`shelfbtemp`. One model owns the ice–ocean interface. The `-ocean given` machinery existed all along but silently fell back to `th` because its input is a FESOM1.x-era bundle FESOM2 never wrote; `fesom2ice` now synthesizes it (plus three exhumations of never-run 2D-path code: `3f07d7a2`, `74671c27`, `bb9d00c2`).

1.3 Bug II — the exchange weights scrambled the coupled ocean (2026-07-14)

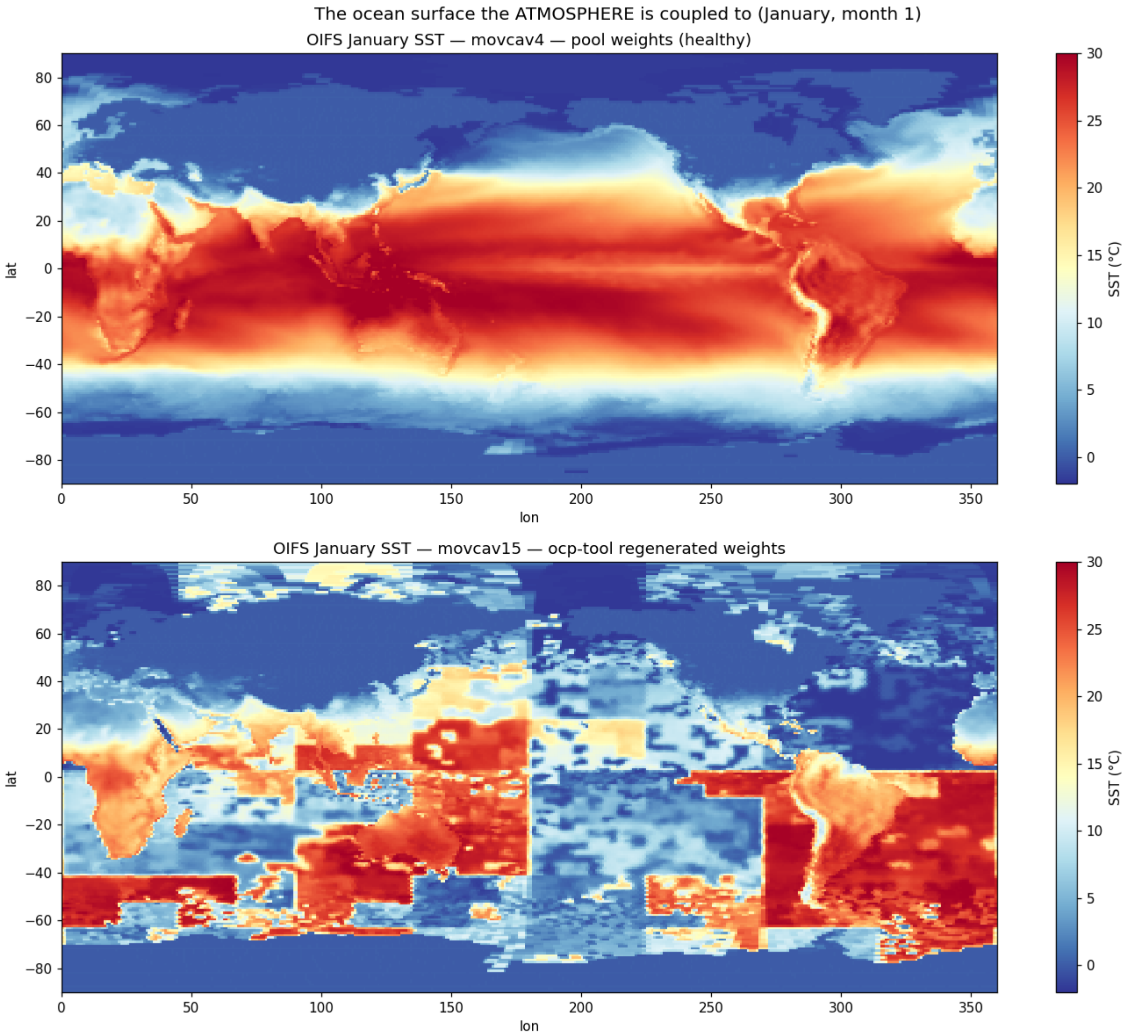

Figure 2: **The picture that settles it.** January SST as OIFS holds it. Top: pool weights — textbook. Bottom: offline-engine weights — a *tile-translocated jigsaw*: tropical blocks at 55°S, polar blocks on the equator, a seam at lon 180 where the node ordering breaks.

FESOM exchanges coupling fields in *partition (my_ list/rank) order*; the offline weight engine (2026-06-17) built the feom geometry in *nod2d.out order* (median displacement 58°). Every submesh exchange was geographically tile-translocated (Fig. 2): OIFS saw Weddell ice fraction 0.04 while FESOM had 0.95; global sea-ice death followed within months.

Fix (`c21fe7c2`): `permute_feom_to_runtime.py` reorders the OASIS geometry to runtime order (via `dist_<nproc>/rpart.out`); `validate_feom_weights.py` applies every weight file to real latitudes and **aborts the leg** above 1° median error (nod2d-ordered weights fail at 42–58°, correct ones pass at $\leq 0.025°$).

1.4 Bug III — the OASIS coupling restart was compound-scrambled (2026-07-14)

`remap_oasis_restart.py` gathered in nod2d space, but its input and required output are both *run-time*-ordered: the emitted `rstos.nc` matched *neither* ordering (correlation of `sst_feom` with the restart's own surface temperature: +0.15). OIFS's first coupling interval saw 305 K water under polar air — a one-hour global jigsaw shock that nondeterministically detonated the convection scheme (identical inputs crashed at hour 19, hour 458, or never).

Fix (`abe56ee2`): un-permute via the old mesh's `rpart.out` → gather → re-permute via the new mesh's; correlation now +0.995. Pool `rstos.nc` rebuilt.

1.5 The wave model's frozen coastline (2026-07-15)

Chunk-3 died deterministically at its first physics step in `voskin_mod.F90:282` squaring WAM's Stokes drift. WAM has no runtime tiling: its sea-point set is baked into `wam_grid_tables` at preprocessing time and had never been regenerated. A cell that is ocean per the regenerated lsm but land per WAM never receives wave data; the Stokes/Charnock fields are srf-restart-carried, so a *warm* start after a coastline change reads never-written storage (cold starts zero-initialize — why chunk-1 tolerates its 12 mismatch cells and historical lsm changes never hit this). Chunk-3 had 125 mismatch cells. WAM itself was separately exonerated for the Bug-IV crash family (a WAM-off run crashed identically).

Fix: one-time maximal-ocean tables (option A; tolerance patching rejected). WAM's sea-point set was rebuilt from ETOPO1 with the southern limit extended 78 → 85°S, then *max-ocean edited*: every cell the FESOM max-mesh can make wet — open ocean or sub-shelf cavity — is forced to sea at its FESOM bottom depth (only ever adding ocean). 1277 cells opened (29 555 → 30 832 sea points), 283 of them south of 55°S: exactly the band a moving coastline walks through (Fig. 3). The tables never change again, wave restarts stay index-compatible across all future mesh changes, and WAM cold-starts once at the switch leg. Known cosmetic debt: the max-ocean edit did not update the obstruction coefficients.

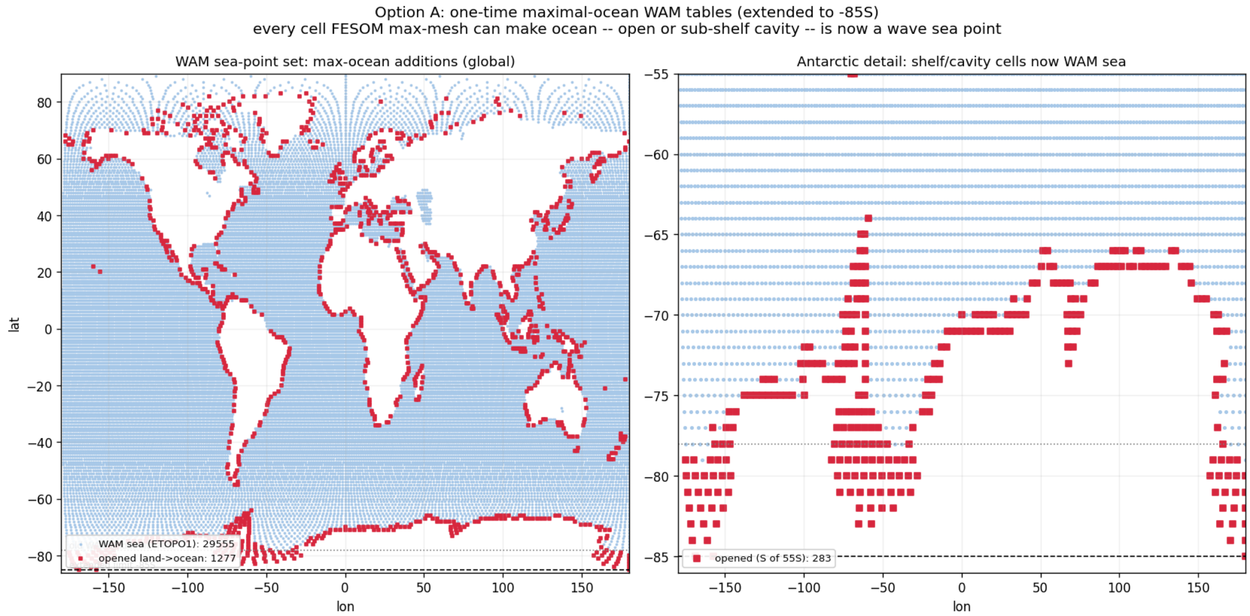

Figure 3: One-time maximal-ocean WAM tables (extended to 85°S). Blue: wave sea points from ETOPO1. Red: cells opened land→ocean because the FESOM max-mesh has a water column there. Pre-registering them closes the WAM-Stokes route into `voskin:282`. (It does *not* close the line entirely: `voskin:282` later recurred through a different, in-step route — §1.19.)

1.6 Bug IV — the ice-surface-temperature coupling is architecturally unstable (2026-07-14→16)

This is the mid-run OIFS crash family (`cubasen`/`vsurf`/`voskin` floating overflow, random timing, polar/MIZ ranks). It survived three partial fixes before the root was found; the full chain is worth recording because each layer was real, just not sufficient.

Layer 1: MIZ aliasing of the sent temperature. FESOM sent raw `ist`; ice-free nodes send the seawater freezing point, so the GAUSWGT cell blend in the marginal ice zone was dominated by warm open water: OIFS's ice tile saw a surface tens of K too warm and returned a strongly negative flux to the fully-iced nodes. Partial fix: concentration-weighted ist (`ist`·`a_ice` ÷ ci on ingest, the EC-Earth/NEMO convention behind `ECE_CPL_NEMO_WEIGHTED_ICE`). Runs survived 3× longer, still crashed.

Layer 2: the FESIM solve has no flux feedback within the coupling interval. FESIM's skin solve relaxes toward $t^* = T_f + \mathrm{a2ihf} \cdot \mathrm{zsniced}/\mathrm{con}$ with the received flux *frozen* for the hour and no $\mathrm{d}Q/\mathrm{d}T$ term: under insulation the equilibrium runs away (measured floors 145–183 K; excursions to −954 K in one run). Partial fix: an implicit flux linearization anchored at the transmitted temperature, $Q(t) = \mathrm{a2ihf} + \lambda\,(t_{\mathrm{ref}} - t)$ with $\lambda = 4\varepsilon\sigma t_{\mathrm{ref}}^3 + \lambda_{\mathrm{turb}} \approx 20\,\mathrm{W/m^2/K}$ (the NEMO-family fallback derivative), energy-closed by booking the linearization heat to the ice growth budget (fesom branch `awiesm3-implicit-ice-surftemp`, `38a558b5`). Dips became bounded and self-recovering in most runs — but not all.

Layer 3 (the root): the atmosphere's ice skin was never solved at all. Send-side instrumentation of `A_Q_ice` (term decomposition + tile state at every $|Q| > 1\,\mathrm{kW/m^2}$ event) delivered the verdict:

- The received flux at cold FESOM nodes is **unphysical and anti-braking**: mean $-15.8\,\mathrm{kW/m^2}$ below 150 K (100% negative), −4.5 kW at 150–180 K, recurring all year with the MIZ seasonal cycle — at **49°N on 0.1–0.5 m ice with thin snow** (Okhotsk / St. Lawrence). Insulation is irrelevant there; no realistic λ brakes kilowatts.
- At the OIFS send, the generating cells have ice-tile fractions down to 0.04 and **tile skin temperatures of ∼204 K under ∼275 K maritime air**; the sensible + latent terms then reach $+17.8\,\mathrm{kW/m^2}$ per tile area. With weighted-ist the tile skins were *still* 189–204 K at substantial tiles — the cold temperature is genuine FESIM state, faithfully communicated.
- The reason the skin can sit 80 K below the air: `vlamsk_mod.F90`, sea-ice tile, with `LNEMOSICOUP=.true.` (the default): `PLAMSK = 1e10` — **the skin conductivity is set to infinity, pinning the tile skin to the prescribed (remapped FESOM) ice temperature**. The atmosphere's surface scheme was forbidden from solving its ice surface; it computed fluxes against whatever arrived through the remap.

The loop, in one line: FESIM's oscillating solve → remap → *pinned* OIFS skin → kW fluxes against an unresolvable surface → remap → FESIM slams further — a two-model positive feedback closed through the marginal ice zone, metastable enough that identical configurations coin-flipped between completing a year and dying at day 2 (runs movcav25–31: raw/weighted × $\lambda = 3.4/20$ × dirty/clean initial state — every combination crash-prone).

Root fix: the coupled slab (§1.8). Stop prescribing the atmosphere's ice skin altogether; let OIFS solve it implicitly each timestep with conduction through the *coupled* FESOM ice and snow thickness. This removes the loop by construction while keeping the flux honest about the actual ice state. Upstream note: mainline AWI-CM3 carries the same pinned-skin architecture and therefore the same latent (if rarer) crash mode.

1.7 Bug V — iterative coupling never handed over to PISM (2026-07-16)

The awiesm3→PISM model switch is resolved at SimulationSetup init by reading the `chunk_date` file, whose path `esm_runscripts/chunky_parts.py` builds by *raw string concatenation of `general.base_dir` before esm_parser resolves variables.* A base_dir containing `${user}` was probed verbatim; the silent `isfile()` miss fell back to "chunk 1, model 1" — awiesm3 self-resubmitted to its own final date and PISM never ran. Proven by log comparison: identical resubmission command lines resolve to `Valid Model Names = ['pism']` under a literal base_dir (movcav16 era) and to `['fesom','oifs',...]` under `${user}`.

Fix (`d61297e8`): `_early_resolve()` expands `${...}` against `general` (environment fallback) in all three chunk-file path builders. Regression-tested against a real chunk file, and validated live: the first genuine awiesm3→PISM handover of the campaign (movcav27), and a full awiesm3→PISM→couple_out→chunk-3-couple_in chain (movcav29).

1.8 The coupled-slab architecture (2026-07-16/17)

The new division of labour: **OIFS owns the ice surface temperature; FESOM owns the ice state.** OIFS's surface scheme solves the ice-tile skin implicitly within every atmosphere timestep (full radiative + turbulent $\mathrm{d}Q/\mathrm{d}T$, finite skin conductivity), with slab conduction through the coupled per-ice thickness and snow depth. FESOM's albedo still drives the radiation; FESIM's own ist solve remains for its internal state (albedo, melt ponds) but is read by nobody — the feedback loop is cut by construction. The machinery is EC-Earth/KNMI heritage (`ICESTATENEMO`, 2008; the SI3 branch of `callpar`) — our contribution is wiring the FESOM branch, which lacked only the thickness.

New exchange field. FESOM sends `sit_feom` = effective (grid-mean) ice thickness `m_ice` (o2a collection, `A_Ice_thickness`); `ECE_FESIM_GET_ICE_STATE` divides by the received ice fraction (standard ε-guard) for the per-ice thickness the slab needs, falling back to the climatological 1.5 m only where the cell has no significant ice. Snow thickness (`snt_feom`) was already exchanged and now actually reaches `surfseb` via `SURFL%ZSNTICE`.

The switch set (verified in code, not inferred):

switch	setting	verified effect
`LNEMOSICOUP`	**.false.**	`vlamsk`: sea-ice tile skin conductivity becomes *finite* (land-tile-style lookup) instead of 10^{10} — the skin is genuinely solved by the SEB. The prior default pinned it to the prescribed ice temperature; this pin was the kilowatt engine of Bug IV.
`LNEMOLIMTEMP`	**.false.**	disables the hourly overwrite of OIFS's 4-layer ice temperature (`SP_SB YTL`) from the FESOM ist ingest; the profile is OIFS-prognostic (ICMGG-initialized, restart-carried). FESOM's ist send becomes inert.
`LNEMOLIMTHK`	**.true.**	activates the `callpar` fill of `ZTHKICE/ZSNTICE` from the coupled fields — the skin/top-layer conduction feels the real ice thickness and snow insulation. (The deep 4-layer discretization keeps its fixed depths; the skin conduction is the term that controls winter fluxes.)
`LMCCDYNSEAICE`	.true.	master enable; false force-resets the whole `LNEMOLIM*` family (`sumcc`).
`LNEMOLIMALB`	.true. (default)	FESOM's albedo drives OIFS radiation (`surfrad_layer` → FESIM getter; raw–raw convention, consistent).
`ECE_CPL_NEMO_WEIGHTED_ICE`	.true. (inert)	only acted inside the disabled ist ingest.
FESOM `latm_owns_ist`	**.true.**	FESIM growth budget takes the atmosphere flux directly (`Qatmice=-a2ihf`, the ECHAM convention); qres/qcon/qlam booking off.

The `sumcc` guard "`LNEMOSICOUP` without `LNEMOLIMTEMP` does not make sense" correctly enforces that a pinned skin requires a prescribed temperature; our mode flips both off. In one sentence: *before, OIFS's ice surface was a mirror bolted to whatever FESOM's oscillating solve sent; now it is a real solved surface conducting through FESOM's actual ice and snow.*

Validated (movcav33). The first coupled-slab run completed its year crash-free and handed over to PISM. With the instrumentation still armed: kW-scale send events at real ice tiles went from 1410–1930 per run to **zero** (117 residual events are all phantom slivers with ice fraction $\sim 10^{-7}$, a `ZMASK` threshold cosmetic); event tile skins moved from 189–212 K into the physical 247–265 K range (Fig. 4); the fluxes FESOM receives at its coldest nodes are $-63 \ldots -80\,\mathrm{W/m^2}$ (textbook polar values, was $-10\,\mathrm{kW}$), and FESIM's internal temperature tracks its anchor to $< 0.1\,\mathrm{K}$.

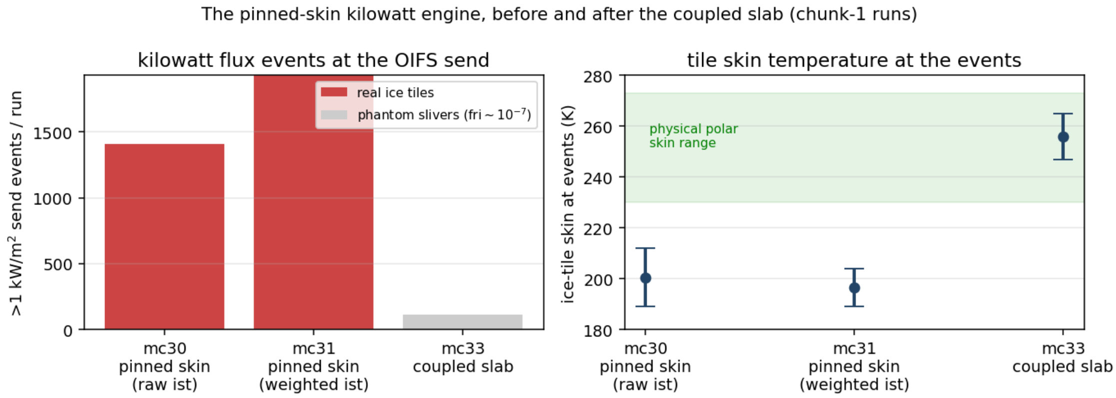

Figure 4: The kilowatt engine, before and after. Left: >1 kW/m^2 events at the OIFS send per chunk-1 run. Right: the ice-tile skin temperature at those events — pinned-skin runs sit 60–80 K below the physical range; the solved skin cannot.

1.9 First fruits of the clean decade: cavity melt against observations (2026-07-17)

The chunk-1 year that forced the first clean PISM decade, split by sector and set against the satellite record (Rignot et al. 2013; Adusumilli et al. 2020):

sector	observed (m ice/yr)	movcav33 chunk-1 (m w.e./yr) mean	median	nodes
Filchner–Ronne	0.3–0.4	0.89	**0.20**	1085
Ross	~0.1	0.90	**0.31**	1075
Amundsen–Bellingshausen	14–27 (PIG 16)	7.8	6.4	229
Antarctic Peninsula	0.4–3.8	7.5	4.4	186
Wilkes (Totten sector)	4.7–9.1	8.3	6.4	296
Amery	0.6–0.7	4.0	3.3	128
E. Weddell / Dronning Maud	0.3–0.9	4.4	3.5	287
circum-Antarctic	0.8 ± 0.7	2.84	0.79	3286

The cold/warm dichotomy that dominates the observations is reproduced: the cold-cavity giants sit at 0.2–0.3 m/yr medians while the warm sectors run 4–8, and 13% of the cavity area refreezes (marine ice, as observed under Ronne/Amery). Residual biases: the East Antarctic coastal sectors (Dronning Maud, Amery) run ~5× warm, and the Amundsen runs ~2× cold — both worth revisiting once multi-year coupled statistics exist. Caveats: node statistics (unweighted), m w.e. vs the observations' m ice (×0.92), one coupled year, and PISM's initial geometry carries reduced versions of the big shelves. For scale: the jigsaw-era coupling produced 28 m/yr.

1.10 The increment test crashes, twice — and passes (2026-07-17/18)

The first chunk-3 leg — awiesm3 on the PISM-changed 208 438-node mesh, stage 6/6 — absorbs the mesh change and runs for weeks of model time before FESOM dies of an SSH explosion (η jumping ±40 m in one step at the Antarctic Peninsula tip; day 19–51 depending on MPI reduction order — the state is metastable). The autopsy was cheap for two reasons: the leg costs 3–4 minutes of wall time, and a temporary XIOS file group with `sync_freq="1d"` flushes daily fields to disk as they complete, so even a crashed run leaves full evidence (default XIOS files sync only on close: 51 days of output, zero records). The hunt found *two independent bugs*, stacked at the same place.

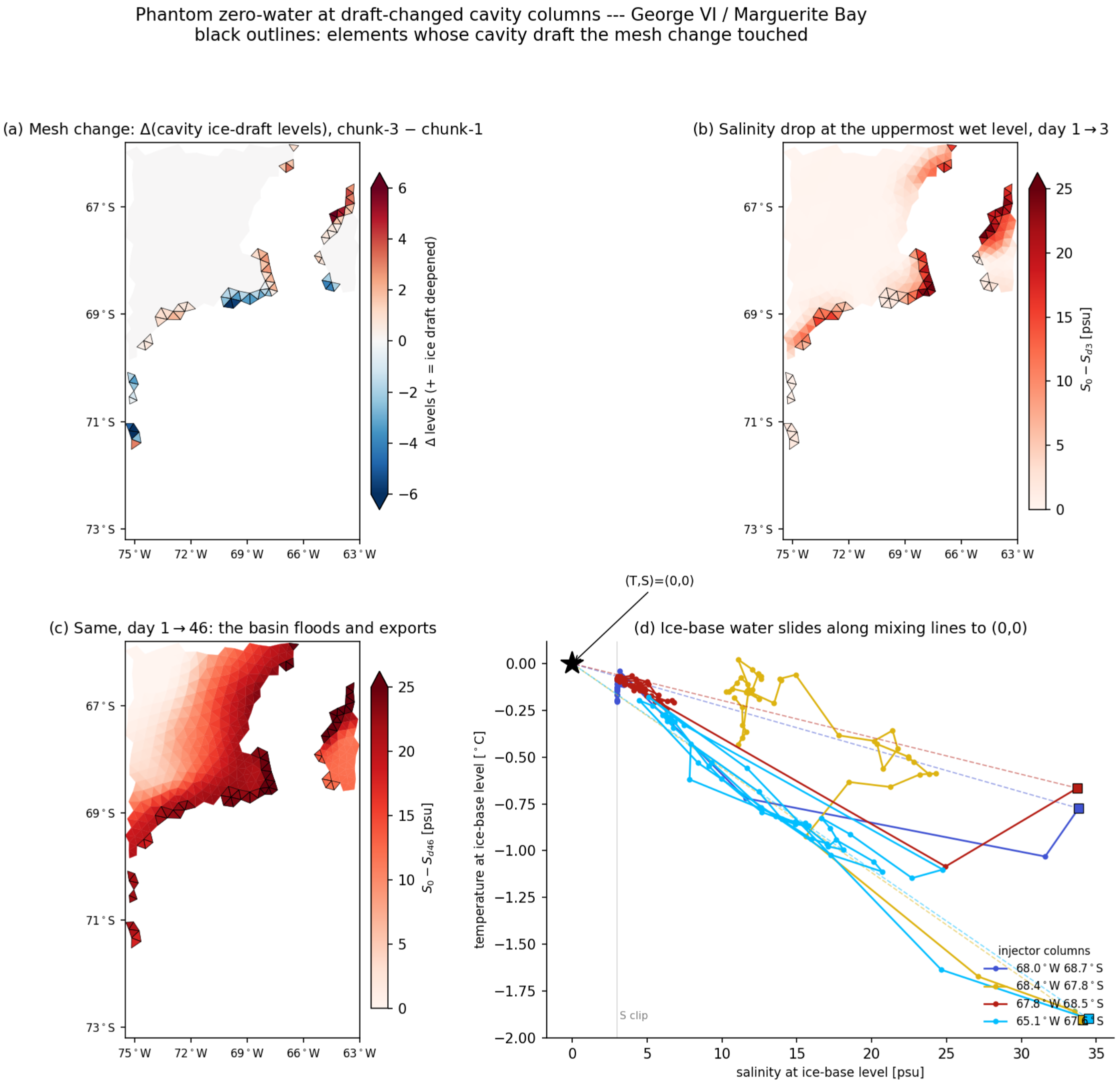

Figure 5: Bug VI seen in one basin (George VI / Marguerite Bay; black outlines = the 73 elements whose cavity draft the mesh change touched). (a) The mesh change: ±1–6 cavity levels. (b) By day 3 the uppermost-wet-level salinity has dropped up to 25 psu *exactly on the outlined elements.* (c) By day 46 the basin has flooded and exports along the coast. (d) The fingerprint: (S, T) at the ice-base level slides along the mixing lines toward $(0, 0)$ with *identical* T- and S-replacement fractions — not a flux (surface freshwater is 50× too small and negative), but a *multiplicative drain*: both tracers lose the same fraction of their value every step.

Bug VI — the layer-thickness mismatch drain

The remapped restart carries `hnode` inherited from the old mesh's zstar state (e.g. 9.964 m where a formerly ice-free column scaled its layers with η). At runtime, cavity columns and their open-ocean edge ring (`ulevels_nod2D_max>1`) keep layer thicknesses *fixed at the mesh nominal* (10.000 m) and are excluded from the per-step `hnode`↔`hnode_new` sync — so the mismatch is immortal. FESOM's tracer T* update applies values · (`hnode` − `hnode_new`)/`hnode_new` every step: a −0.36%/step multiplicative drain on T *and* S. Per-operator probes at a Getz injector column measured −0.1226 psu/step against a predicted 33.64 × 0.0036 = −0.1224; an unchanged FRIS control column measured exactly zero. At free-surface nodes next to the cavity edge the drain even *accelerates*: η grows, the nominal-vs-restart gap grows with it. **Fix** (FESOM, `restart_thickness_ale`): after the restart read, force `hnode` at the whole excluded class back to the fixed nominal and rebuild the depth arrays. Validated by intervention: the drain vanished (+0.0002/step), day-1 fields healthy.

Bug VII — the drowned island

With Bug VI fixed the run survived longer but still exploded at the Peninsula tip, η climbing a steady +0.11 m/day. The reason is geometric, not numeric (Figs. 7, 6): the ice2fesom regrid remapped PISM's *categorical* mask **bilinearly** — a few-cell island surrounded by mask-4 ocean smears to ≈3.x, rounds to ocean, and is deleted from the submesh. PISM's own bed there is +38 to +218 m above sea level: bedrock island, regardless of ice. (The island entered the campaign *bare* — the spinup state carries 0.0 m of ice on all ten cells; a thin 0.1–0.4 m cap has been *growing* during the coupled decades. A second, independent deletion path was found while examining it: PISM encodes ice-free bedrock as mask=0, which the node classifier silently defaulted to ocean; fixed — bare bedrock above sea level is land.) The ocean then drives a runaway coastal jet where a coastline should be. **Fix** (ice2fesom): remap the categorical mask nearest-neighbour; rebuild the mask=5 outside-domain marker (which `remapnn` extrapolation would erase) from a bilinear domain indicator. Continuous fields stay bilinear.

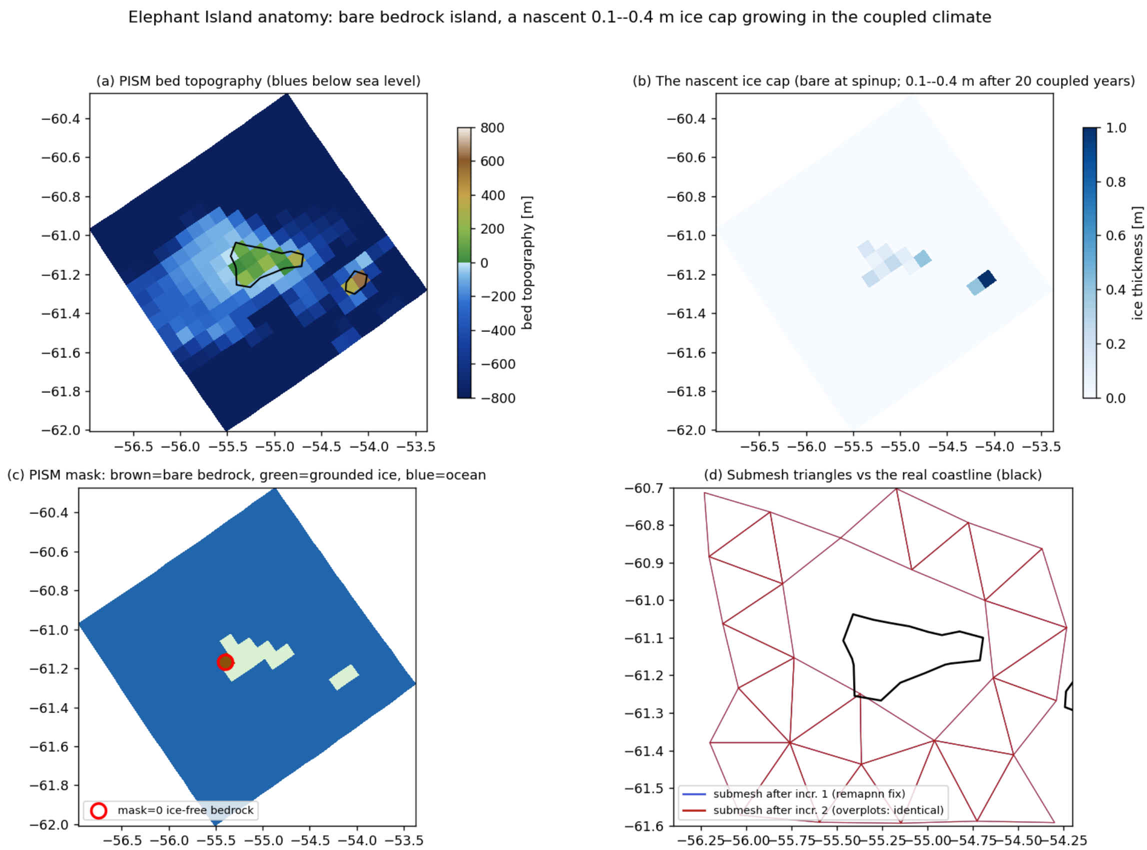

Figure 6: Elephant Island anatomy. (a) PISM bed topography, ETOPO-style colors: a genuine bedrock island (+4 to +297 m) on its shallow platform, Clarence's peak to the southeast. (b) The *nascent* ice cap: bare at spinup, 0.1–0.4 m grown over 20 coupled years (DEBM accumulates here). (c) The PISM mask; the red circle is the single mask=0 (ice-free bedrock) cell that exposed the classifier hole. (d) The submesh renders the island identically after increments 1 and 2 — the nearest-neighbour fix holds.

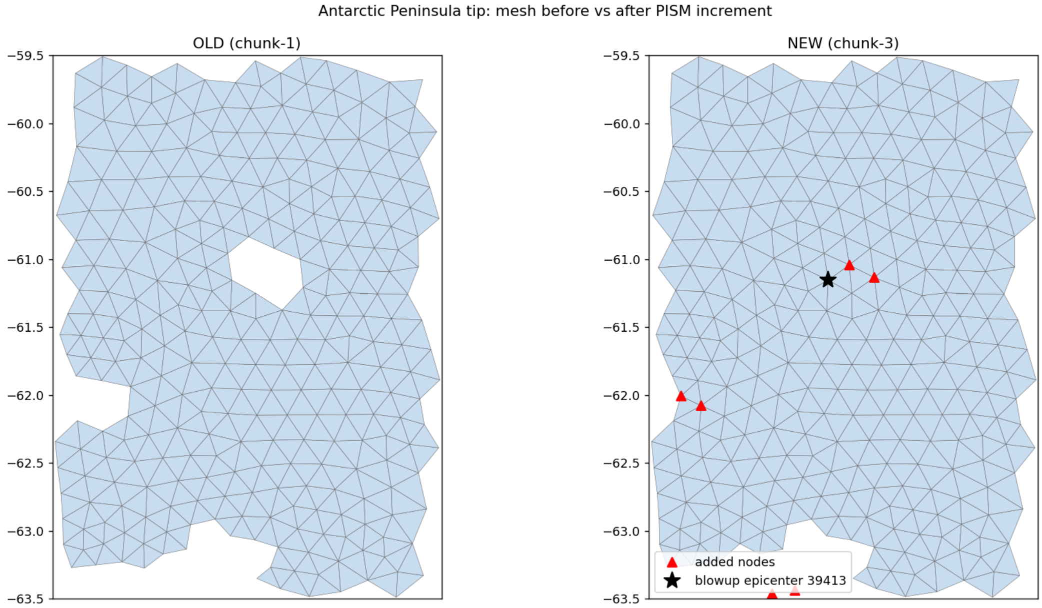

Figure 7: Bug VII. Left: the chunk-1 mesh has Elephant Island (the hole at 55.4°W, 61.1°S) and the Clarence/Gibbs notch. Right: after the PISM decade the bilinear mask remap has rounded both islands to ocean — the added nodes (red) fill them in, and the blowup epicenter (star) sits exactly on the drowned island.

Ruled out by measurement (each was a plausible suspect): surface freshwater fluxes; OIFS runoff/calving and the `ECE_LANDICE` snow cap; the OASIS `feom` masks; the remapped T/S state (freezing cap ran: 1137 columns); mesh-file level-range consistency (zero violations); the zero-filled dry restart levels (padding them — kept as a permanent hygiene pass — reproduced the crash bitwise); and the vertical-advection top boundary (cavity guards in both the implicit and explicit schemes changed nothing — see §4). The couple_in warning about a missing `cap_new_cavity_temp.py` is a red herring: the freezing cap runs as Fortran inside the remap binary.

The increment test passes

With both fixes, movcav33 chunk-3 completed the full year 1901 on the changed mesh (25 min wall, no blowup; the Peninsula node's SSH wanders seasonally instead of pumping) and handed over to the chunk-4 PISM decade unattended — **stage 6/6 closed**. The long run movcav34 (10 awiesm3 years / 100 PISM years) then repeated chunks 1–4 cleanly on the *minimal* build (hnode fix only; the advection guards were reverted after being falsified). Fixes are committed: FESOM `awiesm3-implicit-ice-surftemp`, esm_tools `development_interactiv_mesh_awiesm3`.

The next problem: the second increment and the hygiene overreach (2026-07-18)

movcav34's chunk-5 (year 1902, after 20 PISM years) blew up at day 16 at the western Ross front, where the second increment opened cavity columns wholesale (ulev 13 → 1) amid a 506-node ice advance. The new coastline there was structurally rotten: 24 nodes within 1.5° of the epicenter held by ≤3 elements, including an 18-levels-under-ice cavity column held by exactly two triangles. A *coastline-hygiene sweep* was added to the submesh tool (iteratively remove ice-domain nodes held by ≤2 elements) and the chain then ran unattended through year 1911 and 120 PISM years — eleven further increments without a FESOM blowup, ending only in a recurrence of the old OIFS `voskin` overflow at chunk 13.

The overreach. At wrap-up the evolution diagnostics exposed the truth (Figs. 8, 9): the hygiene sweep had *cascaded through the narrow cavity bands* — fringing shelves and cavity margins are exactly the few-element-wide structures the criterion eats rim-inward — and every post-1902 submesh carries only 14–111 cavity nodes instead of ∼2600. The node classifier was healthy (2223 shelf nodes flagged); the lake floodfill removed only 35 elements; the sweep itself did the damage. **The eleven "stable" legs ran with the ice-shelf cavities amputated — stability by amputation, and every result from movcav34 after chunk 4 is TAINTED.** The sweep was reverted the same day (commit `ca0641b2` reverts `680daa55`).

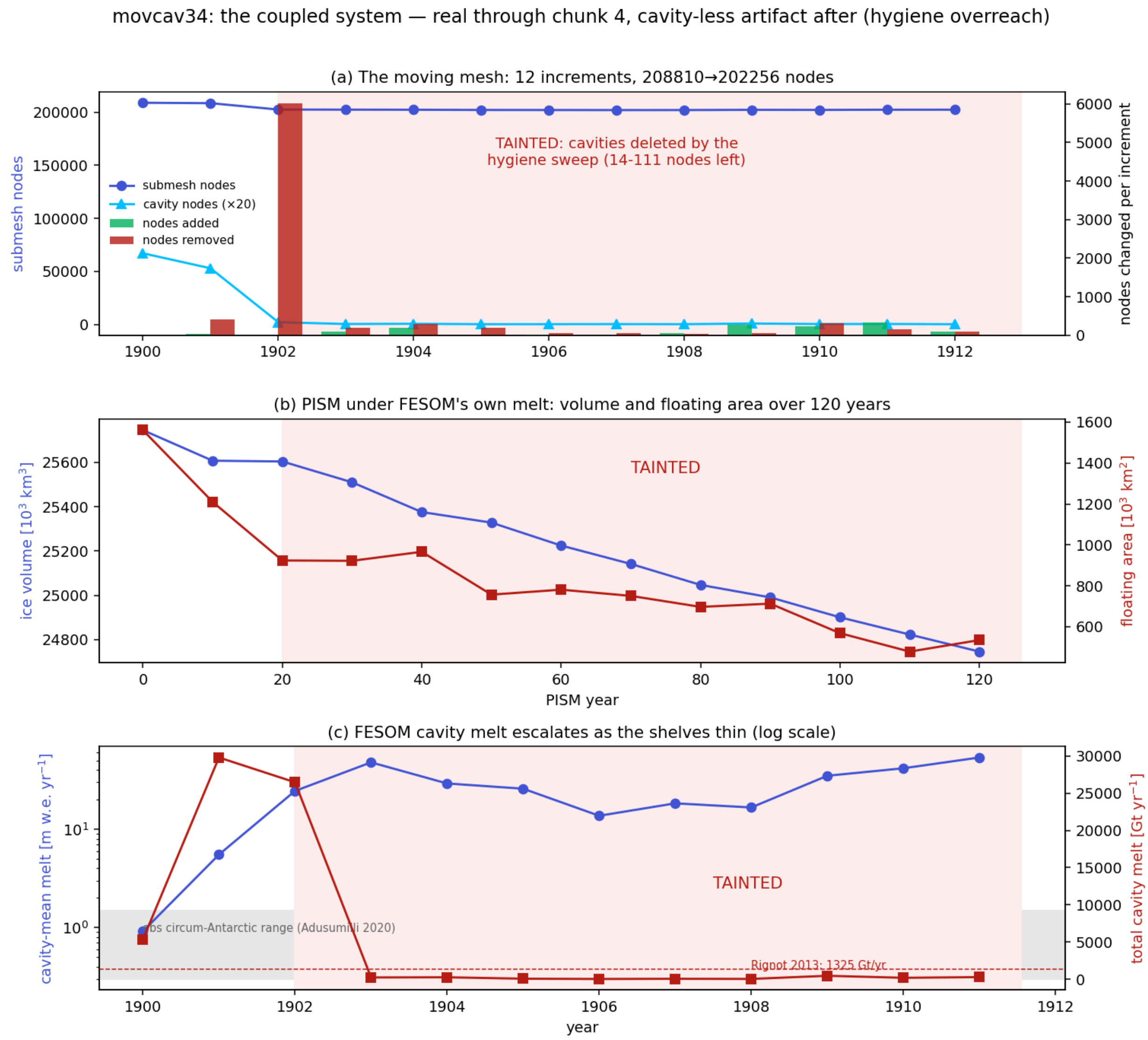

Figure 8: The movcav34 production run, honest version. (a) Submesh and cavity-node counts per increment: the collapse from 2647 to 111 cavity nodes at the 1902 mesh is the hygiene amputation. (b) PISM volume and floating area over 120 years: the decline is real coupled melt through year 20, then continues under forcing computed from an essentially cavity-less ocean. (c) Cavity-mean and total melt per leg (log scale): the escalation beyond ∼1902 samples only the few surviving cavity slivers. Red shading = tainted.

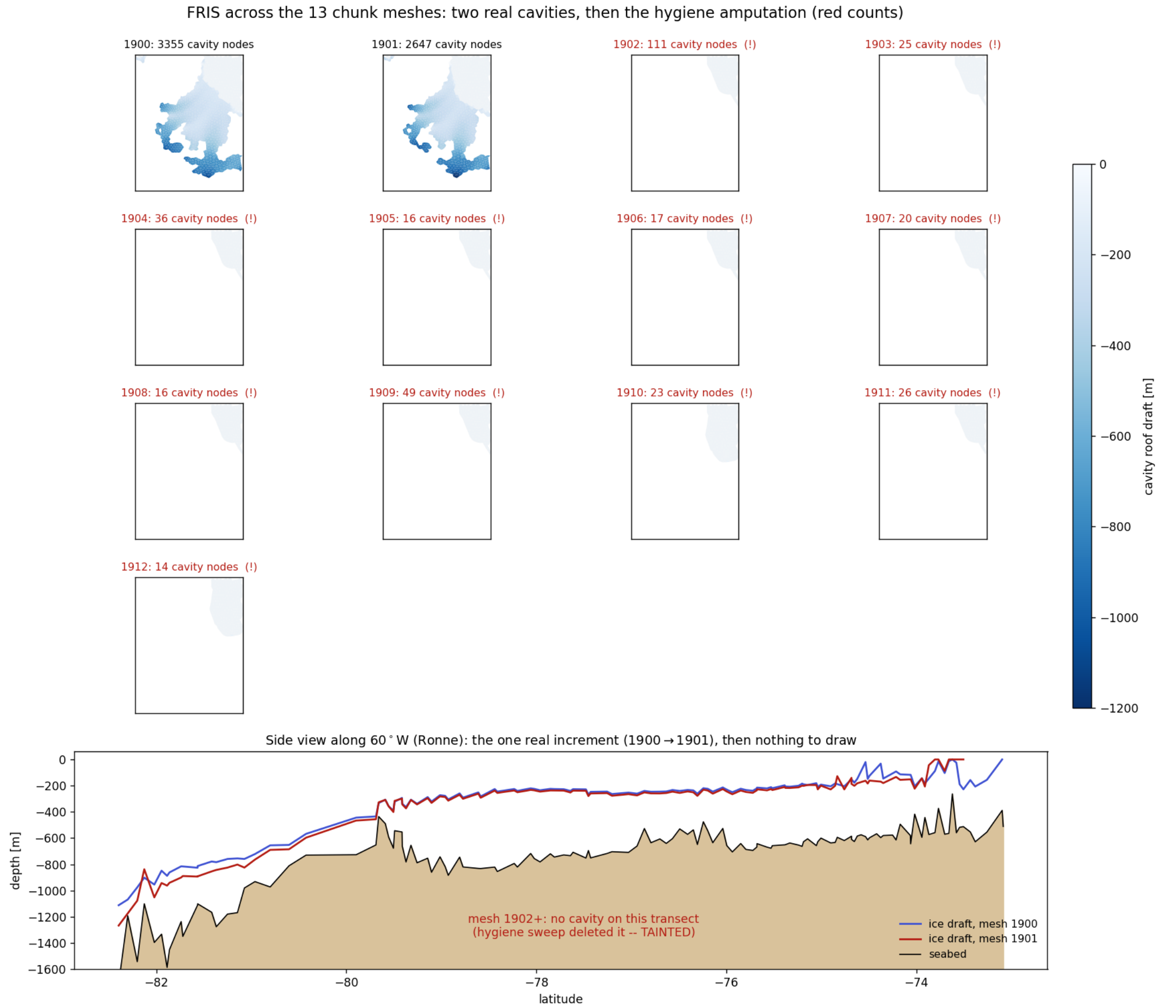

Figure 9: FRIS across the thirteen chunk meshes. Top grid: cavity elements coloured by roof draft — two real cavities (1900, 1901), then eleven empty frames with red node counts. Bottom: ice draft along 60°W for the one genuine increment (1900→1901; the southern draft deepens by 50–100 m), after which there is nothing left to draw.

The problem to solve next (Figs. 8, 9 are its statement): make the mesh increments survivable *without* destroying the cavities. What is genuinely proven and stays: the full cycle awiesm3 → PISM → awiesm3-on-a-changed-mesh works end-to-end with real cavities (movcav33 chunk-3 + handover; movcav34 chunks 1–4 reproduce it on the minimal build). The work queue, in order: (i) a cavity-preserving fragile-sliver treatment — removal must not disconnect or cascade through the few-element-wide cavity bands (guard on the shelf flag, a connectivity-preserving criterion, plus a safety valve that aborts the leg if the submesh cavity count drops far below the classifier's flag count); (ii) rewind movcav34 to chunk-5 and re-run the increments with real cavities; (iii) the recurring OIFS `voskin` coastal overflow; (iv) the iterative-coupling loop ignores `final_date`; (v) the escalating warm-bias melt that drives PISM's fast geometry increments in the first place. Items (i) and (v) are worked below.

1.11 The coastline is not the problem — the increment is (2026-07-18)

The Ross-front blowup looked like a fragile coastline (24 nodes held by ≤ 3 elements within 1.5° of the epicentre, including an 18-level cavity sliver on two triangles), so the first response was a hygiene sweep removing ice-domain nodes held by ≤ 2 elements. It ran the chain to year 1911 — but by *amputating the cavities* (Bug "VIII", above). An offline harness on the saved 1902 flag files then settled the question by building submesh variants without submitting jobs:

- **Fragility is normal, not pathological.** The meshes that ran clean full years carry the *same* fragile census as the crashed one: pool 3470 nodes with ≤ 2 elements (429 in cavities), submesh_1901 3448 (378), versus 1902's 3444. A control that kills the hypothesis: if $\leq$ 2-element slivers were fatal, 1901 would have died too.
- **The increment is the violence.** Measured as per-node cavity-roof change against the previous submesh: increment 1 (pool→1901) max |Δdraft| 451 m, mean 0.96 m; increment 2 (1901→1902) max **1196 m**, mean **8.7 m** — 9× the mean, opening 13 vertical levels at the epicentre in one decade beside a 506-node grounding advance. This is downstream of item (v): the warm-bias melt escalates (cavity-mean 0.9→5.5→24 m/yr), so each decade's geometry step is larger than the last.

Fixes committed. (a) A *cavity-amputation abort guard*: if the reduced submesh retains < 50% of the classifier's flagged cavity nodes, the leg fails loudly instead of running a formally-valid but cavity-less ocean (the tripwire that would have caught Bug VIII in one couple_in instead of eleven tainted chunks). (b) The PISM mask-value `0` (`MASK_ICE_FREE_BEDROCK`) is now classified *land*: a second, independent island-deletion path (bare bedrock defaulting to ocean) that would have re-drowned Elephant Island once its nascent cap finishes melting. **Rejected:** a per-increment cavity-roof *rate limiter* (clamp the roof to within 75 m of the previous leg). It worked offline (retained 2221 cavity nodes vs 1864 unclamped) but was *rejected on physics grounds*: it makes the ocean's cavity geometry deliberately lag the ice sheet's, a coupler-imposed inconsistency between components. The increment must be tamed at its source (the melt, next) or by numerical treatment of the post-increment transient — not by distorting the coupled geometry.

1.12 Why the cavity melt is high and rising, and how to slow it (2026-07-18)

Item (v), the disease under the increments: the PI-control cavity melt runs at ∼5.5 m/yr area-averaged against an observed circum-Antarctic ∼0.8 m/yr, and *escalates* (0.9→5.5 m/yr over 1900→1901 on real cavities). Diagnosed from the monthly `temp`/`salt`/`fw`/`unod`/`vnod` output (Fig. 10); the cause is not where the first pass guessed.

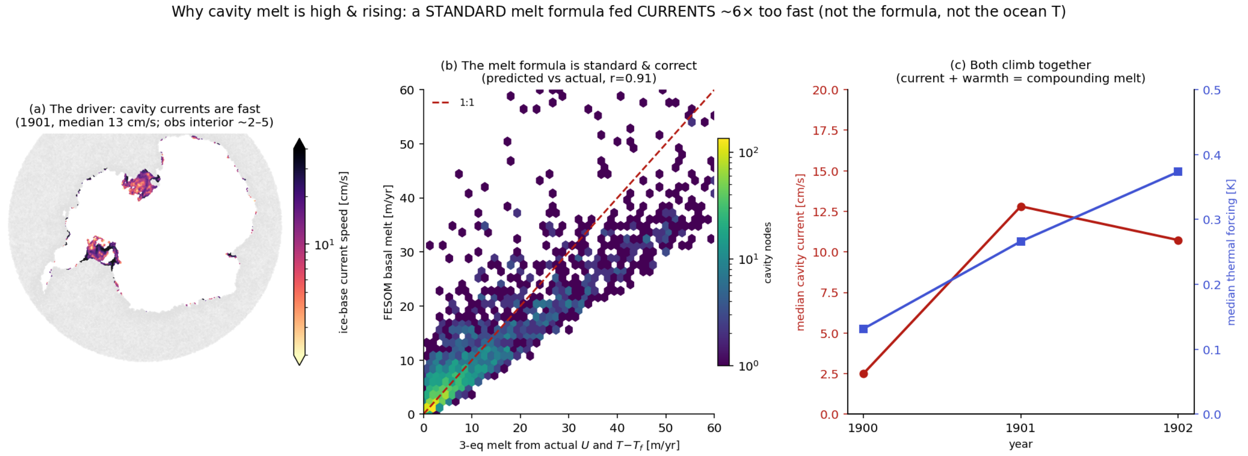

Figure 10: Why cavity melt is high and rising. (a) Cavity ice-base current speed (1901): median 13 cm/s, mean 18, 61% of nodes above 10 cm/s — fast, concentrated at the Amundsen/Bellingshausen shelves and the Ross/Filchner ice fronts (observed cavity interiors are a few cm/s). (b) The melt *formula* is standard and correct: fed FESOM's actual currents and thermal forcing, the three-equation scheme reproduces the model's own melt (predicted vs actual $r = 0.91$). (c) Both drivers climb together — thermal forcing (median $+0.13 \to +0.37$ K, warm-water intrusion) and current speed — so the melt compounds.

The chain, measured. Thermal forcing $T - T_f$ at the ice-base level is *near-realistic* (area-weighted +0.14 K; median +0.13 in 1900 rising to +0.37 in 1902). The Holland & Jenkins three-equation scheme in `cavity_param.F90` is standard (drag $a_k = 2.5 \times 10^{-3}$, $\gamma_T \approx 10^{-4}$ m/s), and fed the model's own currents and thermal forcing it *reproduces the melt* ($r = 0.91$, 18 vs 16 m/yr): the formula is not miscalibrated. By elimination — melt 7× too high, formula validated, temperature realistic — the excess lives in the **cavity currents**: median 13 cm/s where ∼2 cm/s would give the observed melt, i.e. ∼ 6× too fast. Because the exchange velocity is $\gamma_T \propto |U|$, that fast circulation is exactly what inflates the melt; and as the shelves thin the currents speed up further — the escalation.

How to slow it without killing the climate response. The design constraint is to dissipate *momentum* with a flow-dependent operator, never to cap the diagnostic (velocity, γ_T, or melt) — a quadratic drag still lets the current find a slower equilibrium that shifts with forcing, whereas a cap freezes the response. The ice base is *not* free-slip: `cavity_momentum_fluxes` already applies quadratic drag $-\rho C_d|U|U$ at `nzmin`, correctly ordered (cavity elements skip the atmospheric- stress overwrite, then receive the drag). But it borrows the *seabed* coefficient $C_d = 2.5 \times 10^{-3}$ with the developer's own flag in the source ("*need to check the sensitivity to the drag coefficient*"). The plan: split it into a separate, tunable `C_d_cavity` (default $= C_d$) and raise it — physically justified, since ice-shelf bases are hydraulically rougher than a flat seabed (basal crevasses, terraces, keels), and dedicated cavity models treat ice-base drag as a distinct tunable $\gtrsim$ seabed. Being quadratic it preserves the velocity's climate sensitivity. Complementary flow-dependent levers if drag alone is insufficient: a vertical-viscosity floor in the cavity boundary layer, or Smagorinsky horizontal viscosity in cavities. *Not* to be done: raising the global C_d (over-brakes the open-ocean bottom boundary layer everywhere). Validation target: melt toward ∼0.8 m/yr *while preserving* the spatial melt–forcing pattern ($r = 0.81$) and the warm-Amundsen/cold-FRIS contrast — brake the amplitude, not the geography. Caveat: part of the $2.5 \to 13$ cm/s jump over 1900→1901 is plausibly spin-up plus the increment shock, so retune against a settled multi-year state, not a transient.

On bias correction (a design question raised alongside): observation-anomaly ("delta") ocean forcing is standard for stand-alone ice-sheet runs, and even cavity-resolving projections bias-correct their *atmospheric* forcing (Naughten et al. 2018 force FESOM with CMIP anomalies on an ERA-Interim climatology, control on reanalysis) — but it would not cure this run: the thermal forcing here is already near-realistic, so correcting T/S addresses a problem we mostly do not have. The excess is in the *circulation*, upstream of both the melt formula and the water it is fed.

Superseded (2026-07-19): the "too-fast currents" diagnosis holds, but the proposed cure — a tunable ice-base drag `C_d_cavity` — does not: the standalone dissection below shows the fast currents are *injected by the increment*, not an under-damped equilibrium (chunk-1 and both year-long cold-start equilibria run ∼2.6 cm/s at the stock drag). See §1.13; the drag plan is moved to Falsified.

1.13 The spin-up is the remapped restart, not the water — a standalone dissection (2026-07-19)

To test the "fast currents" hypotheses cheaply, a *standalone* FESOM harness (`build_sa`, submesh_1901, 20-day runs, ∼2 min each) with a cavity median-column-speed metric. It reproduces the disease and then dissects it (Fig. 11).

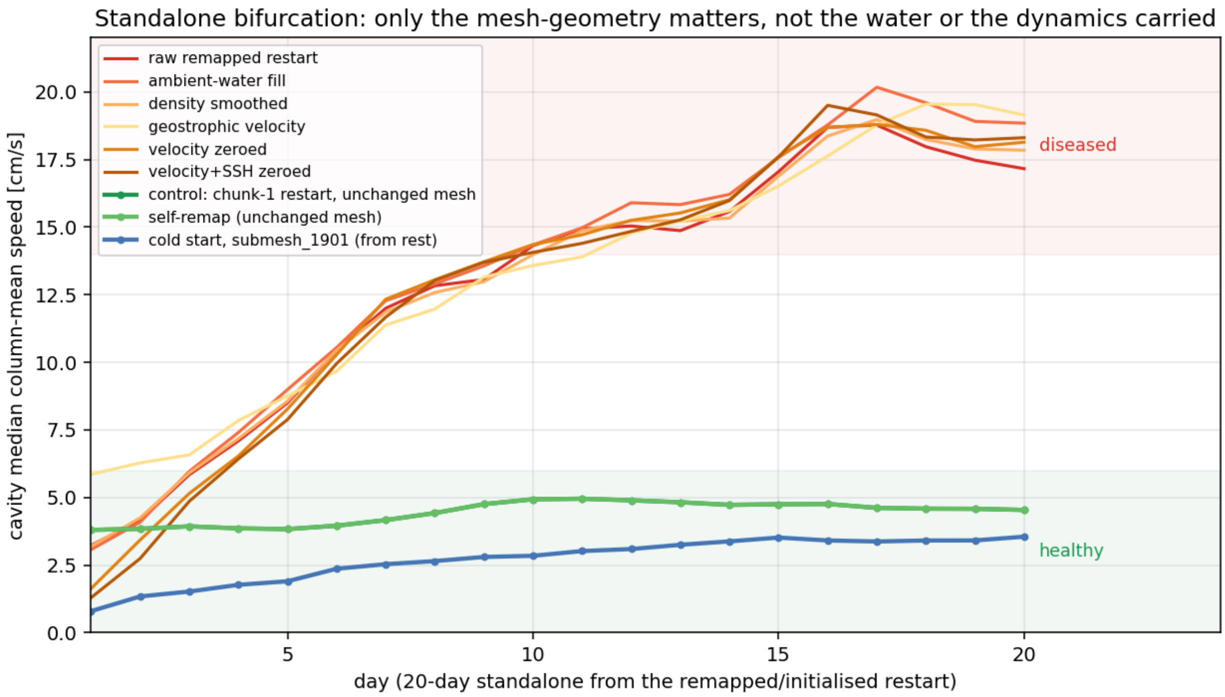

Figure 11: Standalone cavity median speed vs. day. The runs bifurcate cleanly by *what mesh geometry* they use, not by the water or the dynamics carried. **Healthy (∼2–5 cm/s):** chunk-1 restart on the unchanged mesh (control), the same restart remapped onto its *own* mesh (self-remap, exonerating the remap machinery), and a cold start from rest on the changed submesh_1901. **Diseased (climb to ∼17–19 cm/s):** the raw remapped restart — *and* every attempted fix of the carried state on that mesh: cavity density smoothed, cavity refilled with ambient open-ocean water, geostrophic velocity imposed, velocity zeroed, velocity+SSH+hbar zeroed. All fast, all indistinguishable. The excess is density-independent and not fixable in any single carried field.

The elimination. (i) The harness is faithful — unchanged geometry stays slow, through the remap or not. (ii) The cavity *water mass is irrelevant*: chunk-1 water, harmonically smoothed water and a fresh ambient reservoir all spin up identically — baroclinic/density is not the driver. (iii) The *geometry is fine*: a cold start on submesh_1901 from rest settles to ∼2.6 cm/s (1-year run, peaked ∼month 3 then relaxed) — a healthy equilibrium lives on the new mesh. (iv) It is not the individually carried velocity, SSH or hbar (zeroing each → still fast) and `hnode` is already nominal. **Conclusion:** the injector is the *remapped restart being internally inconsistent with the new mesh*;

the geometry *change* is only the trigger that exposes it, not the cause.

Footprint. Matching submesh_1901 to the initial submesh by coordinate: of 2627 cavity columns, **1602 changed vertical structure but the remap flags only 410**. The classifier flags only columns that *gain* wet levels (need fill: ice base shallower or seabed deeper); the **1183 columns whose ice base got *deeper*** (thicker ice above → moved ice-loading/pressure boundary) are carried untouched. A density-independent, geometry-linked lead — not yet a proven cause.

Learning the target (in flight). Two 1-year cold starts with *identical* climatology and forcing, differing only in the mesh (1900 vs 1901), give two directly comparable equilibria. Their difference Δ_{equil} at the changed nodes is the *pure* ocean equilibrium response to the ice-draft change — the first-principles target a one-shot remap rule must reproduce (the old restart ≈ the 1900-mesh equilibrium; the modification should push it to the 1901-mesh equilibrium, derived, not copied).

1.14 The seam field is `hnode` (2026-07-20)

The equilibria then localised the seam to a single field. Building a rest state (velocity, SSH, AB, hbar all zero) with the *equilibrium* density and stepping it still spun up (median 19 cm/s) — so density is finally, unconfoundedly dead. That rest state differs from the healthy cold start in exactly one carried field: `hnode`, the ALE layer thickness. Swapping the remap's `hnode` for the equilibrium's (the *only* file that changes, by ≤ 0.73 m) flips the cavity from 18.6 to 4.5 cm/s, reproduced to two digits (Fig. 12). The changed-column differences sit a couple of levels below the moved ice base — which is why every density/velocity/SSH test walked past it. (*Initially* read as "differences entirely at the changed columns"; the correction below shows the swap also moved the open-ocean `hnode`, which carries most of the effect.)

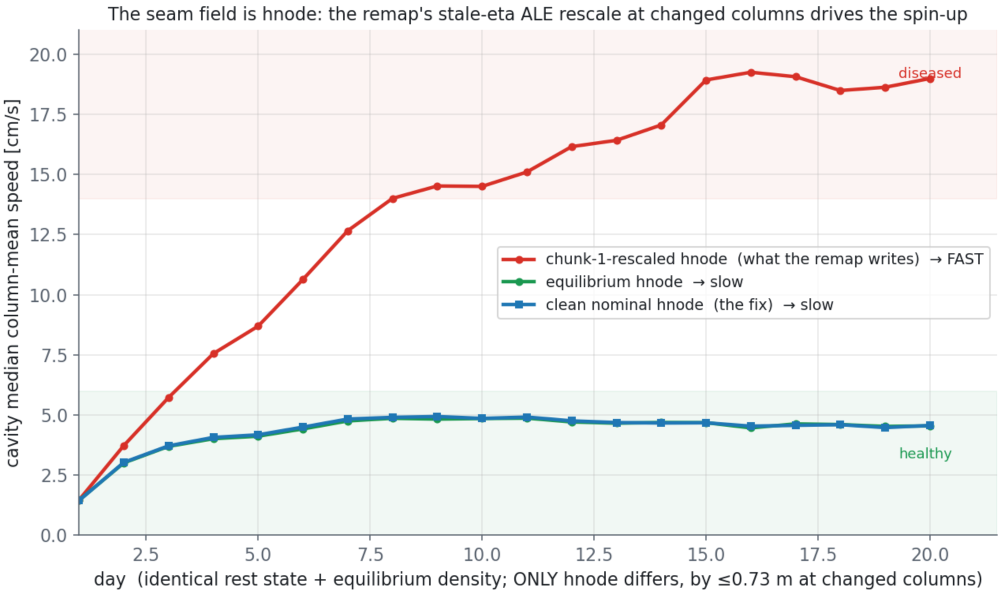

Figure 12: One field, one flip. Identical rest state and equilibrium density; *only* `hnode` differs, by $\leq$ 0.73 m at the changed columns. The remap's chunk-1-rescaled `hnode` (red) spins the cavity to ∼19 cm/s; the equilibrium `hnode` (green) and a *clean nominal* `hnode` *applied everywhere* (blue) both hold the healthy ∼4.5 cm/s. *Caveat (added below):* this swap also changes the *open-ocean* `hnode`, not only the changed columns — so it overstates what a changed-column-only fix achieves (that reaches only ∼10 cm/s).

Mechanism. `remap_hnode` rescales every changed column so $\sum h = (z_{\text{bar}}(ul) - z_{\text{bar}}(nl)) + \eta_{\text{new}}$, using the *remapped chunk-1* η. At changed columns that η is stale — starkest at cavity→open ("closed") nodes, where it is the under-ice depressed value (≈ -1.6 m) applied to a now-open column

— so the rescale *compresses* `hnode` (e.g. 10.0 → 9.19 m). That inconsistent thickness is exactly the "standing dh/dt source at t=0" the rescale was added to prevent; it seeds the barotropic spin-up. The rescale does the opposite of its intent because it trusts the carried η.

Fix, and the correction (2026-07-20 late). The `remap_hnode` edit — nominal z_{bar} thicknesses at the geometry-changed columns (including the ice-base-deepened ones the classifier calls "unchanged") instead of the stale-η rescale — was implemented and regenerated end-to-end. In the standalone harness it *halves* the over-speed, 17 → 10.3 cm/s: a real, physically-clean improvement, but *not* a full restoration. Reaching ∼4 cm/s requires nominal `hnode` over the *whole* ocean: the fully-fixed and the target restarts turn out *identical in the cavity* and differ only in the *open-ocean* `hnode`, and a cavity+ice-front-halo nominalisation does not help (11 cm/s) — only nominalising everywhere does. So the earlier "one field, one flip" was *confounded*: the clean swap carried the open-ocean `hnode` too, not just the changed columns. The residual is not the cavity fill but the **vertical-coordinate seam** at the ice front — zstar stretches the open-ocean layers by the free surface ($h_k = h_k^{\mathrm{nom}}(D+\eta)/D$) while the rigid-lid cavity keeps them nominal; the remapped, zstar-stretched open-ocean `hnode` is what pushes the cavity from 4 to 10. This reframes "nominal everywhere" not as a global-reset artifact but as precisely what `linfs` (fixed layers, η a top-boundary diagnostic) does by construction — removing the seam without any `hnode` surgery. **Confirmed**: the raw remap (chunk-1 `hnode`, no surgery — the exact restart that spins to 17 cm/s under zstar) run with `which_ale='linfs'` does *not* spin up; it decays to ∼1.7 cm/s median / 2.5 mean and holds. So the seam, not the cavity fill, was the mechanism, and `linfs` is the clean complete fix. Status: changed-column `remap_hnode` fix implemented (uncommitted, gives 10.3 — now superseded by `linfs`); the open question is whether `linfs` is acceptable model-wide for the coupled AWI-ESM3 open-ocean climate (a linearised free surface with mass-conservation trade-offs vs. zstar).

1.15 `linfs` in the coupled chain, and the open-cavity fill (2026-07-21)

The standalone finding was then taken into the real iterative coupling: a fresh run (`movcavlinfs`) with `which_ale='linfs'` from chunk 1, everything else identical, 1200 s timestep. It validates the seam story end-to-end (Fig. 13). Four mesh increments (1900–1903) hand over with *no* cavity spin-up and *no* CFL blowup; the cavity node count is *preserved* (∼2668, vs the zstar run's collapse 2647 → 111), the per-increment mesh change *shrinks* as the ice-ocean system settles (402 → 187 nodes added), and cavity melt is stable and realistic — ∼0.22 m w.e. yr^{-1} cavity-mean, ∼1344 Gt yr^{-1} total (Rignot ∼1325), with none of the 0.9 → 5.5 m yr^{-1} escalation seen under zstar. The vertical-coordinate seam was the mechanism; `linfs` removes it in the coupled system too.

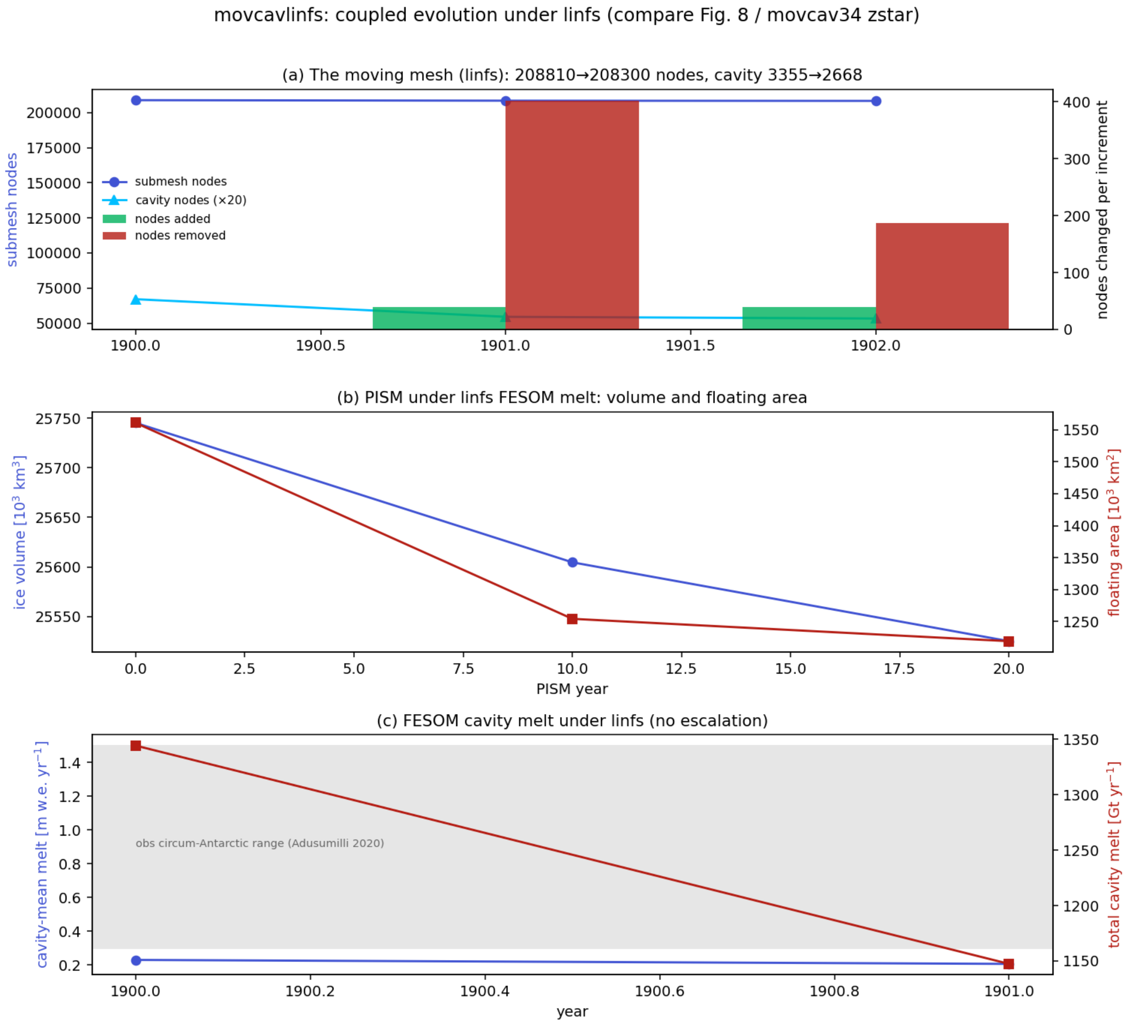

Figure 13: `movcavlinfs` coupled evolution under `linfs` (compare Fig. 8 / `movcav34` zstar). (a) The moving mesh: node and cavity-node counts per increment, with nodes added/removed; the increment shrinks and the cavity is preserved. (b) PISM ice volume and floating area. (c) FESOM cavity-mean and total melt — stable in the observed circum-Antarctic range, no escalation.

The 1904 stop was a fill artifact, not the seam. Chunk 5 (1904) tripped FESOM's fixed temperature sanity reject ("`temperature becomes NaN or <-5.0`") at step 1 on a cavity node, $T = -5.35$. Not CFL, not spin-up. Two nested causes. (i) The remap carries each newly-iced node's *own open-ocean* column into the new cavity, so where the ice front advances over warm water the ice base inherits it — up to +6.6 °C of thermal driving at the new roof (Fig. 14a, red). (ii) A boundary-layer cap (`cap_new_cavity_temp`) that clipped those columns to the local freezing point *over-cooled* the deep part: at ~4500 m the pressure-depressed $T_f = 0.0901 - 0.0575\,S - 7.61{\times}10^{-4}\,z \approx -5.35\,°\mathrm{C}$, dragging real ~ -0.5 °C deep water below FESOM's −5 floor (Fig. 14a, blue).

Fix: seed newly-iced columns from the nearest existing cavity (user proposal). Instead of carrying a node's open-ocean column (then capping it), each open→cavity column is seeded from the nearest column that is cavity in *both* meshes, depth-matched. This gives cold, cavity-appropriate ice-shelf water, regionally correct by nearest-neighbour (Amundsen-warm / FRIS-cold preserved), with the deep column left real — no cap needed. In the 1904 increment it seeds 204 recipients from 2326 donors and cuts ice-base warm columns (> 0 °C) from 61 → 9 and peak thermal driving from +6.6 → +0.8 °C (Fig. 14, green profile / right map); the residual nine are genuine warm East-Antarctic-coast cavities ($\leq +0.8$). A freezing-point floor (clamped to −4.9) is retained only as a last-resort backstop. No cap, no crash, deep water preserved.

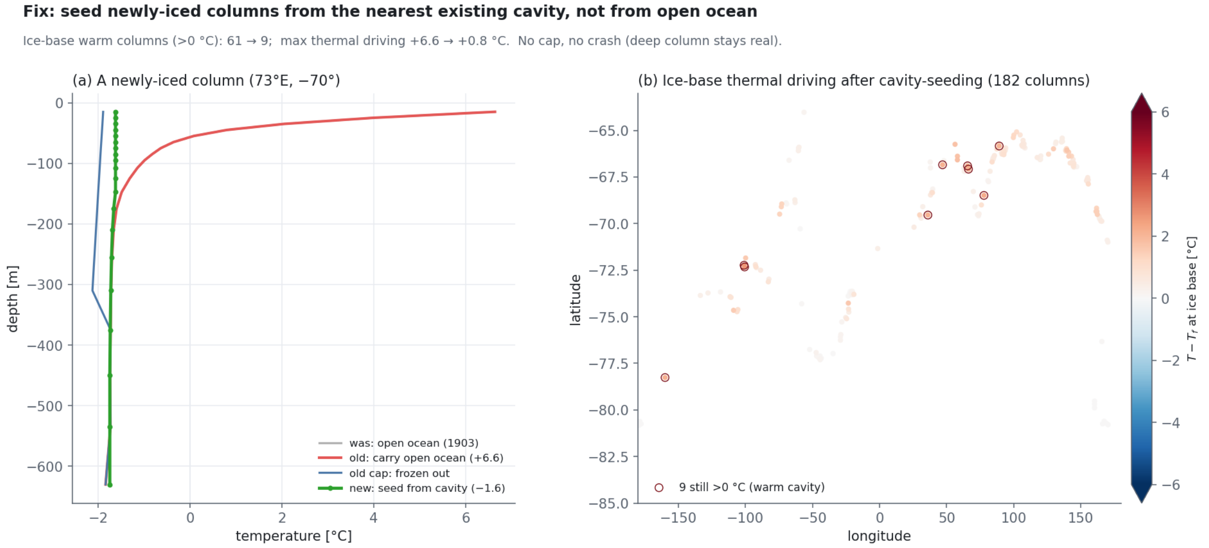

Figure 14: The open-cavity fill fix. (a) A newly-iced column: carrying its open-ocean water (red) puts +6.6 °C at the ice base; the freezing cap (blue) over-cools the deep column; seeding from the nearest existing cavity (green) gives a real cold ice-shelf-water profile with the deep column intact. (b) Ice-base thermal driving $T - T_f$ over the 182 open→cavity columns after cavity-seeding — almost all near freezing; the nine circled ($> 0\,^\circ$C) are physical warm-cavity donors.

1.16 Case closed: the full `movcav37` cycle (2026-07-22)

With both fixes in place — `linfs` (coordinate seam) and cavity-donor seeding (open-cavity fill) — a fresh run `movcav37` was launched from chunk 1 and carried the complete decade 1900–1909: **ten coupled years, nine mesh increments, no spin-up, no blowup** (it passes 1904, where `movcavlinfs` had stopped). This closes the spin-up investigation.

The decisive metric is the cavity current itself (Fig. 15). The area-weighted cavity median column-mean speed holds **1.1–2.1 cm/s across all ten years** (p90 $\leq$ 5.2), never leaving the healthy band — against the 13–17 cm/s the *same* remapped restart reached at every increment under zstar. The barotropic spin-up that this whole report chased is, by construction, gone: it was the vertical-coordinate seam, and `linfs` removes it.

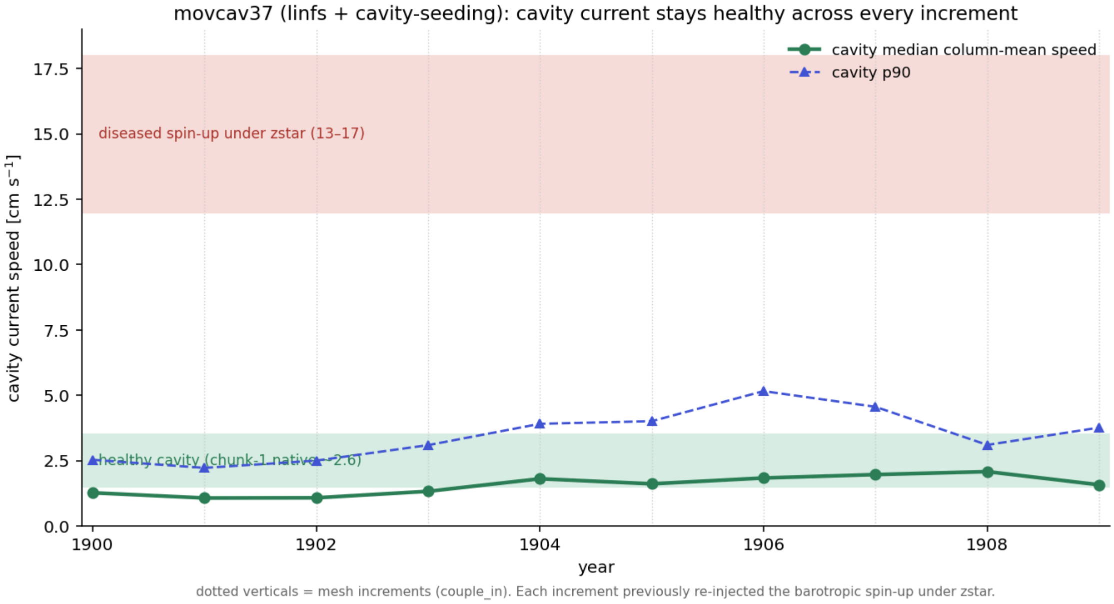

Figure 15: The spin-up, closed. `movcav37` cavity median (and p90) column-mean current speed per year; dotted verticals are the nine mesh increments. The current stays in the healthy band (chunk-1 native ∼2.6 cm/s) at *every* increment — the 13–17 cm/s diseased spin-up under zstar (red band) never appears.

The coupled system is also *settling*, not diverging (Fig. 16). The cavity is preserved (3355→2318 nodes, a gentle thinning, not the zstar collapse to 111), the per-increment mesh change shrinks (405 →∼60 nodes), PISM volume and floating area decline smoothly with no shocks at the increments, and cavity melt stays inside the observed circum-Antarctic range (0.23→0.82 m w.e. yr^{-1}, 1329→1685 Gt yr^{-1}) — no trace of zstar's escalation to 5.5 m yr^{-1}. The one feature to watch on a longer run is the mild *upward* melt trend; the flat 1–2 cm/s currents identify it as thermodynamic/geometric (shelves thinning, cavity warming slightly), not a dynamic spin-up, and it remains within observations.

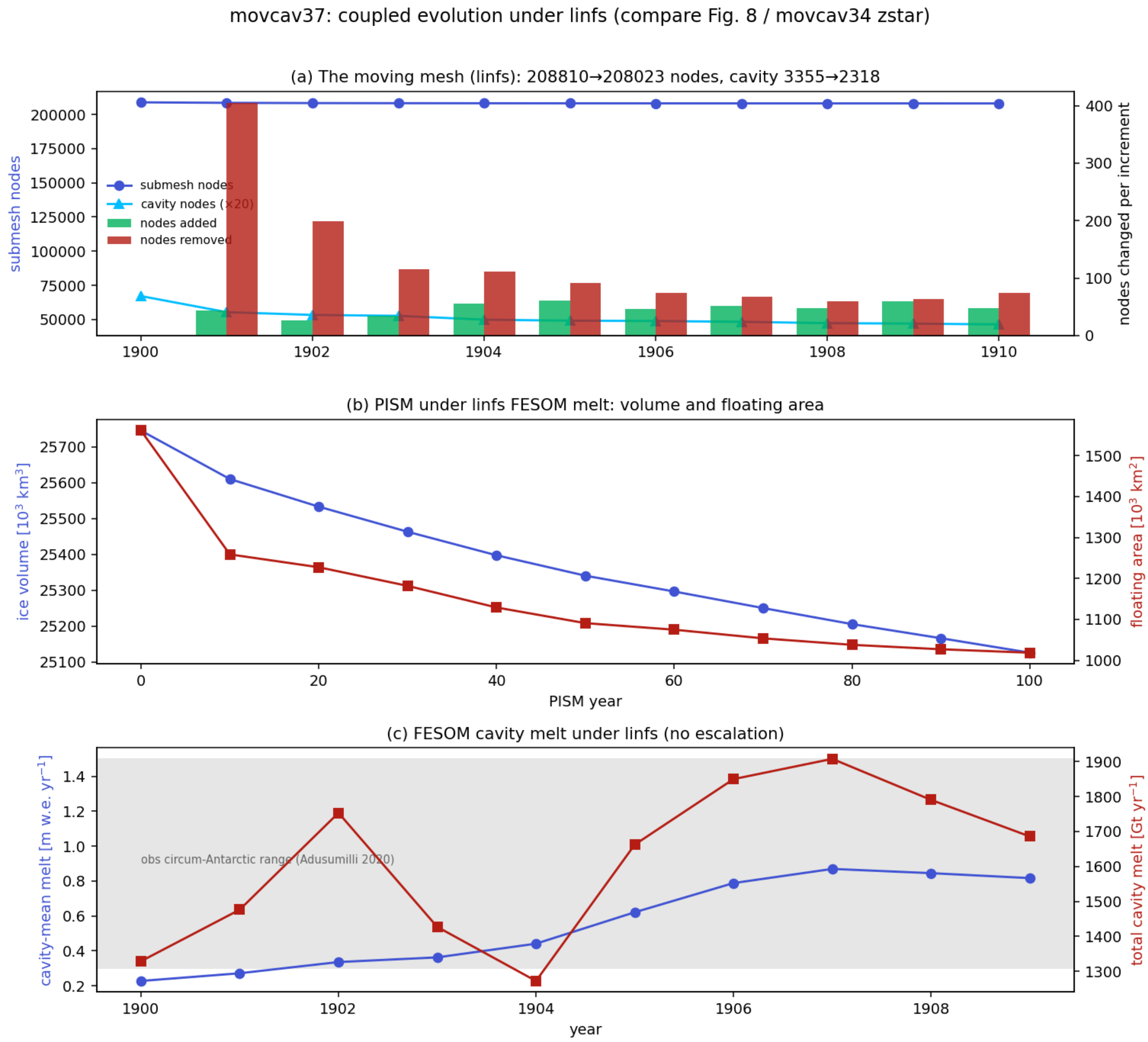

Figure 16: `movcav37` coupled evolution (compare Fig. 13 / `movcavlinfs`, now the full decade). (a) Moving mesh: cavity preserved, increments shrink. (b) PISM volume and floating area, smooth. (c) FESOM cavity melt, in the observed range, no escalation.

1.17 Sea ice and snow in `movcav37`: melt-back restored; the Ross lagoon (2026-07-22)

The open item of §2.2 — the SH summer melt-back and the coastal snow pile-up — is answered by `movcav37`. Diagnosed properly (sea ice lives on *open-ocean* surface nodes; cavity nodes carry spurious values and must be excluded), the thickness fields sit in the observed range in both hemispheres (Figs. 17, 18).

- **SH summer melt-back is restored.** The winter (Sep) pack is a coherent 0.4–1.0 m ring (thicker in the Weddell/Ross gyres); by summer (Feb) it retreats almost completely to thin Weddell/coastal remnants — the strong seasonal melt-back the reference had and the earlier movcav runs lacked. Hemispheric mean thickness ∼0.6 m (obs ASPeCt/ICESat 0.5–1.0 m).
- **The coastal snow pile-up is gone.** SH snow is <0.15 m over the pack with a coastal rim ∼0.25–0.35 m, within the observed 0.1–0.4 m — not the 2–3.5 m band of the pre-fix runs. (The `linfs` + clean chunk-1 ice init + coupled slab combination did restore the melt-back; this closes the §2.2 question.)

- **NH is realistic**: 2–4 m ice against the CAA/Greenland thinning to the marginal seas (obs ∼1.5–3 m), snow 0.15–0.35 m in winter (Warren ∼0.2–0.4 m).

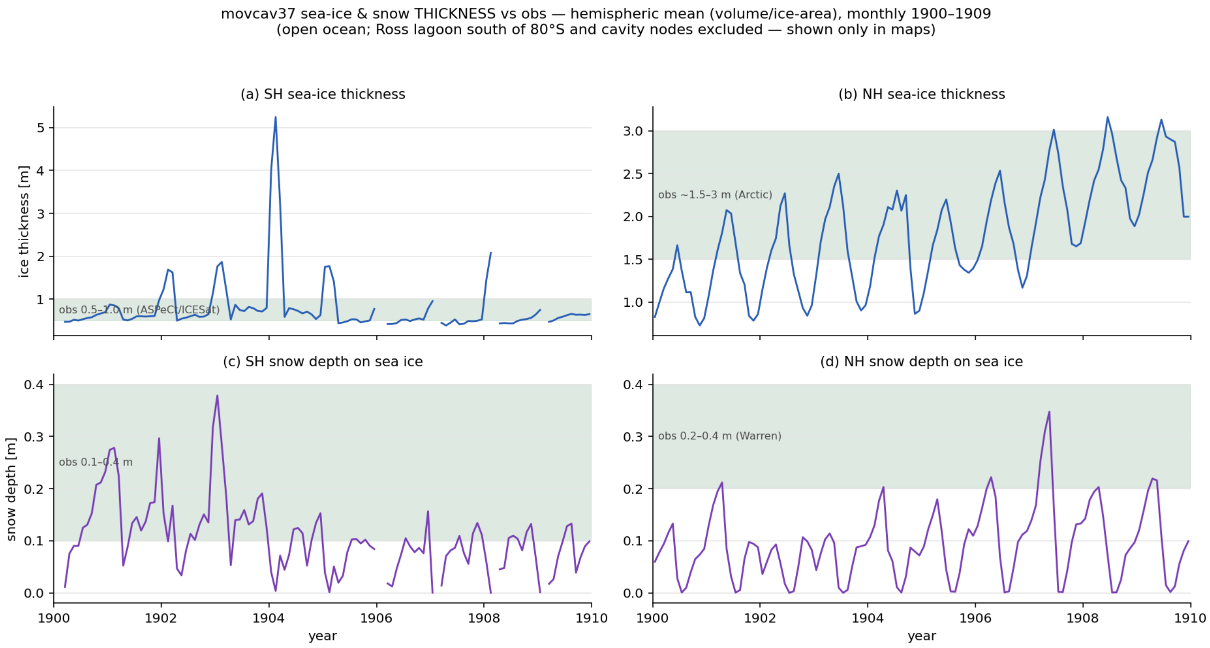

Figure 17: `movcav37` sea-ice and snow thickness vs observations, monthly 1900–1909, hemispheric mean (total volume / total ice-area over ice-covered open ocean, $a_{\text{ice}} > 0.15$). Green bands = observed ranges. Cavity nodes and the Ross lagoon (open ocean south of 80°S) are excluded here and shown only in the maps. SH ice summer spikes are survivor bias (only thick coastal ice remains at the Feb minimum).

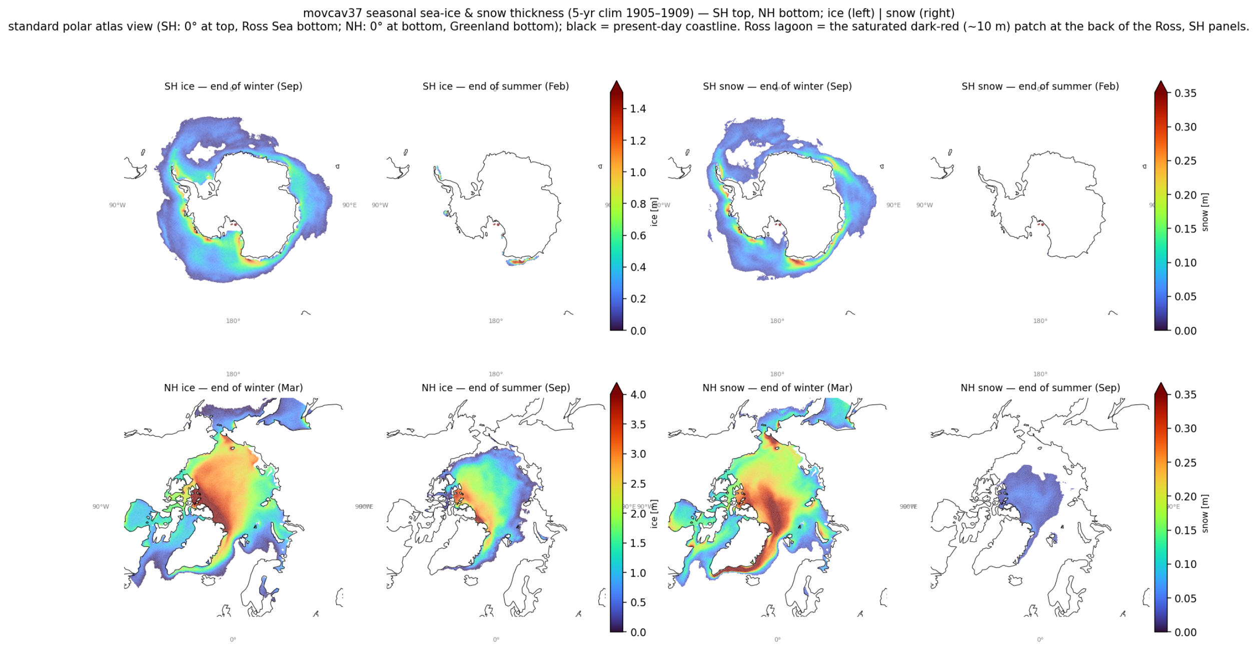

Figure 18: `movcav37` seasonal (end-of-winter / end-of-summer) sea-ice and snow thickness, 5-year climatology 1905–1909, on the native grid (standard polar view, 0° meridian at bottom; grey = present-day coastline). SH top, NH bottom; ice (left pair), snow (right pair). SH ice melts back to near-nothing by February; SH snow is a thin coastal rim. The two dots off the Ross coast are the *Ross lagoon* artifact (below).

New open item — the Ross lagoon. The one unphysical thickness signal left is a small (∼18-node) patch at the back of the Ross Ice Shelf (lon ≈ 180°, lat ≈ −83°) where PISM, over the run, thinned the shelf and retreated the grounding line until the classifier flipped those nodes from *cavity*/*grounded* to *open ocean* (verified node-by-node: `cavity(ul=12-16)` or land in 1900 → `ul=1` by 1909). FESOM then traps thick (∼10 m), convergent, non-melting sea ice in the near-enclosed deep embayment — the dots in Fig. 18. It is a PISM-geometry artifact (likely over-thinning the back of the Ross, fed by excess FESOM cavity melt there), not a sea-ice-model problem; it is excluded from the aggregates above. Fix candidates: guard the classifier against ungrounding interior nodes far behind the shelf front, or address the excess back-of-Ross melt.

1.18 The coupled-slab switch was off for `develop`/`develop-cc`; the sea-ice climate validated on a clean decade (2026-07-23→27)

Rolling the coupled-slab work out from `develop-is` to the sibling setups `develop` and `develop-cc` exposed a silent, months-old configuration bug. The coupled slab is *hardwired in the FESOM build* (§1.8: `ice_thermo_cpl.F90` sets $Q_{\text{atmice}} = -a2ihf$, so the atmosphere owns the ice-surface temperature and its slab conduction must be fed the coupled ice thickness). OIFS is told to match through `NAMMCC`: `LNEMOLIMTHK=.true.` (use the coupled thickness in the surface solve) and `LNEMOLIMTEMP=.false.` (do not overwrite the ice skin from the ocean model each hour). The OIFS defaults (`sumcc.F90`) leave *both* `.false.`. Only `develop-is` set `NAMMCC` — and only because its *runscript* did, by hand. So `develop` and `develop-cc` silently ran with `LNEMOLIMTHK=.false.`: the `sit_feom`→`A_Ice_thickness` field added by the whole coupled-slab rollout was delivered over OASIS and then *discarded*, and the atmosphere conducted through a default slab thickness.

The signature was unmistakable once a clean 10-year `develop` run (`si10y`, TCO95–CORE3, 1990 GHG, fixed full mesh, no cavity) was diagnosed: with the switch off, the Arctic went *completely ice-free every June–October* and the winter pack carried half its proper volume (NH March volume

5.3; §2). Setting the three `NAMMCC` switches turned that into a physical sea-ice climate at the first attempt (Fig. 19): NH March volume $5.3 \rightarrow 27.4 \times 10^3\,\mathrm{km}^3$ (obs $\sim$28–30), NH September extent $0.0 \rightarrow 5.6 \times 10^6\,\mathrm{km}^2$, SH February extent $0.0 \rightarrow 1.7$. The fix now lives in the setup config, not the runscripts (`awiesm3.yaml`, commit `0aca2487`), applied to all three `develop*` versions; tags keep the stock defaults because their component versions predate `sit_feom`.

This does not taint the `movcav` results. The `develop-is` runscripts set `NAMMCC` by hand throughout, so every `movcav` run in this report (§1.8 onward) already had `LNEMOLIMTHK=.true.`. The bug lived only in the fixed-mesh siblings.

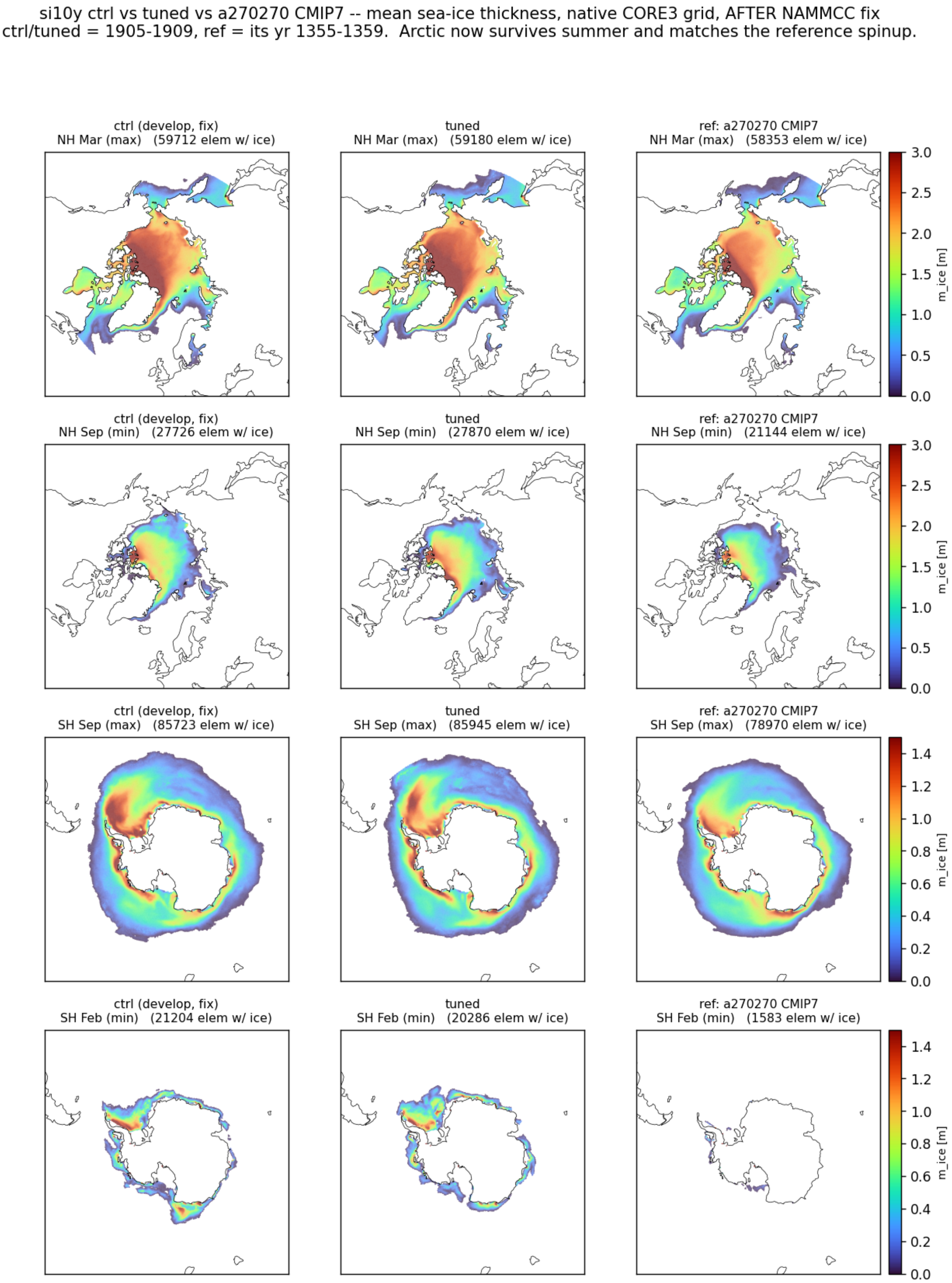

Figure 19: `develop` sea-ice thickness after the `NAMMCC` coupled-slab fix, native CORE3 grid, 5-yr mean of spin-up years 6–10, zoomed to the polar caps. Columns: control (fixed `develop`), the tuning experiment (§2), and an independent 500-year CMIP7 pre-industrial spin-up (a270270) at the same spin-up age. Rows: NH March (max), NH September (min), SH September (max), SH February (min); element counts with $a_{\text{ice}} > 0.15$ annotated. The NH September row — blank ("NO ICE") before the fix — now carries a coherent central Arctic pack *exceeding* the reference (27.7k vs 21.1k elements). The large-scale fields match the reference spin-up hemisphere-by-hemisphere; the coupled-slab config is correct.

Quantitatively, against 1990s observations and the a270270 pre-industrial spin-up sampled at the *same* spin-up age (years 6–10 of each), last-5-year means:

metric	`develop`+fix (1990)	a270270 (PI)	obs
NH March extent [10^6 km^2]	16.1	15.4	∼15.5 (NSIDC 1990s)
NH September extent [10^6 km^2]	5.6	4.7	∼7.0
NH March volume [10^3 km^3]	27.1	23.1	∼28–30 (PIOMAS)
NH March mean thickness [m]	1.8	1.6	∼2.3–2.6
SH September extent [10^6 km^2]	21.4	20.3	∼18.5 (NSIDC)
SH February extent [10^6 km^2]	1.7	0.0	∼3.0
SH September mean thickness [m]	0.7	0.6	∼0.9–1.2

The fixed run is closer to observations than the reference on every row (Fig. 20). Two residual biases remain, both ordinary sea-ice tuning territory and neither addressed by the `whichEVP`/melt-pond tuning that was tried (it moved nothing): **NH ice is ∼20–25% too thin** (thin pack, under-done September minimum), and **the SH seasonal cycle is too extreme** (over-grows in winter, over-melts in summer). A single-point thickness runaway (∼52 m at year 10) is present *identically in the reference* (which reaches 855 m by its year 500): a shared FESOM point-convergence artifact, not introduced by the coupling.

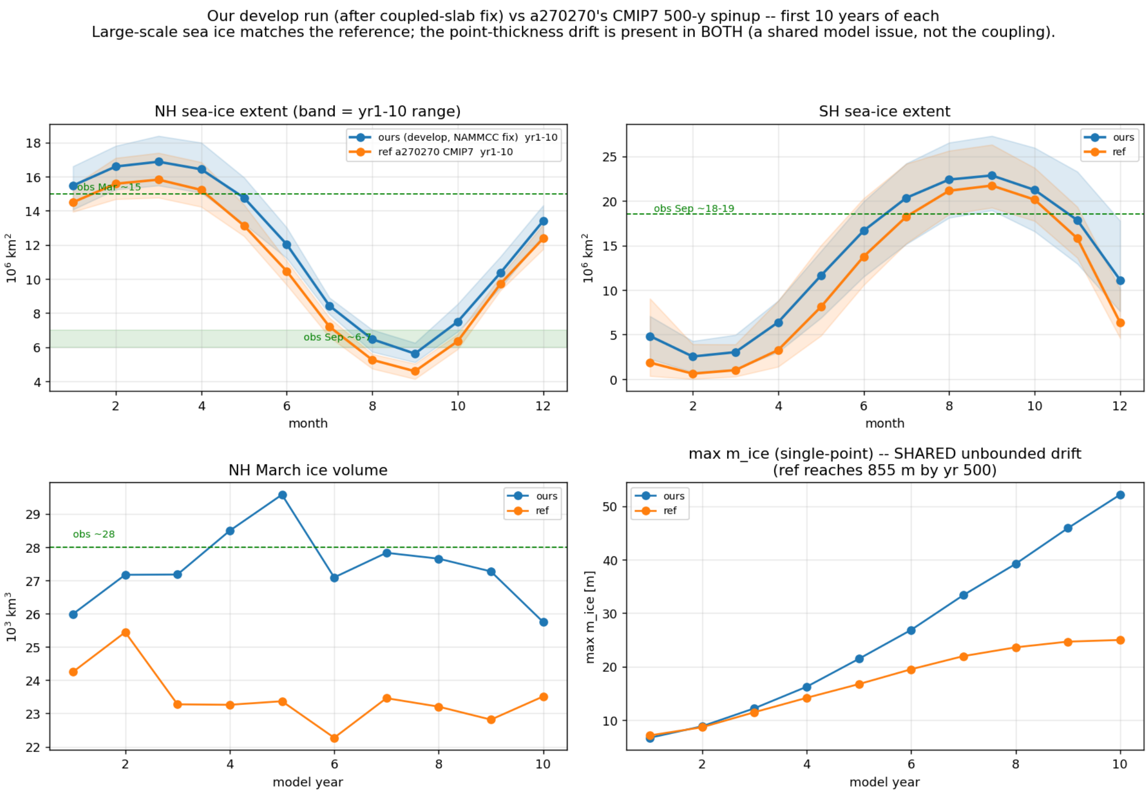

Figure 20: `develop`+fix (blue) vs a270270's CMIP7 pre-industrial spin-up (orange), first 10 years of each. (a,b) NH/SH seasonal extent, band = year-range; green markers = observed. (c) NH March volume; (d) single-point max thickness — the unbounded drift is present in *both* runs (shared model artifact). Large-scale sea ice matches the reference; where the two differ, the fixed run sits closer to observations.

A throughput trap surfaced while setting these runs up, worth recording because no short test catches it: with `oasis3mct.lresume=false` the coupling restart is rebuilt every exchange and the step rate decays ∼100× over a run (0.28 → 11.4 s/step); `lresume=true` (from a matching restart set carrying `sit_feom`) holds it flat at ∼0.10 s/step. See Method notes.

1.19 The `voskin:282` overflow, inventoried across the campaign and instrumented (2026-07-24)

The WAM fix (§1.5) closed one route into `voskin_mod.F90:282`, but the line kept killing later legs. Sweeping every `error (72)` floating overflow in the `develop-is` history gives eleven crashes, all on OIFS ranks and only ever at the rank extremes (OpenIFS decomposes north→south, so the extremes are the poles — never the mid-latitude ranks):

run(s)	chunk	hemisphere	crash site
movcav18/24/26/28/30/31	1 (1900)	both poles	`cubasen:453`, `vsurf_mod:280`
movcav19	5 (1904)	Arctic	`cubasen:453`
movcav33	2 (1901)	Antarctic	`callpar:678`
movcav34	13 (1912)	Antarctic	`voskin_mod:282`
movcav39	2 (1901)	Antarctic	`voskin_mod:282`
par03	2 (1901)	Antarctic	`voskin_mod:282`

Two families separate cleanly by chunk. **Chunk 1** crashes (`cubasen`/`vsurf`, both poles) are the pinned-skin sea-ice-thermodynamics instability, Bug IV — gone under the coupled slab. **Chunk** ≥ 2 crashes (`voskin`/`callpar`, *Antarctic only*) are the mesh-increment family, and they persist.

`voskin:282` is the *detector, not the cause*. The build is single precision (`libsurf.SP.so`, max $\sim 3.4 \times 10^{38}$); line 282 is the Langmuir number $Z_{\mathrm{LA}} = [\rho_a/\rho_w \cdot Z_{\mathrm{UST}}^2/\max(u_{\mathrm{St}}^2 + v_{\mathrm{St}}^2, 10^{-8})]^{1/4}$, whose argument is $\sim 1.2 \times 10^5 \cdot Z_{\mathrm{UST}}^2$ — it overflows only once the friction velocity Z_{UST} already exceeds $\sim 5 \times 10^{12}$. Upstream (lines 193–204) Z_{UST} is built from the surface stress `PUSTR/PVSTR` and a buoyancy amplifier $Z_{\mathrm{WST2}}/Z_{\mathrm{DU2}}$ that divides by the skin temperature `PTSKM1M`; it is floored by `ZEPUST` but *never capped*. For par03 the crashing leg's inputs were all checked and clean: the seven A096 fields of `rstos.nc` (including `A_CurX/Y`, which set the wind-relative-to-current stress) all physical, the regenerated ICMGG `skt`/`sst`/`ci` physical, and the `A_SST` coastline sanitizer (§1.4 lineage, commit `e5d4796c`) *ran* on that leg yet the crash still occurred. So the bad value is *generated inside the first timestep*, which is why file inspection has not found it.

Two diagnostic probes were added to `voskin_mod.F90` (dump the column inputs when $Z_{\mathrm{UST}} > 10$ or non-finite, and when the buoyancy Z_{BUO} blows up or `PTSKM1M`$< 100\,\mathrm{K}$), OIFS rebuilt (`develop-is`; `VOSKIN-DIAG` strings verified in `libsurf.SP.so`). The instrumented reproduction — par03 chunk 2 crashes $\sim 60\,\mathrm{s}$ in — is **ready but not yet run** (it mutates the par03 experiment). Its output will separate the two candidate routes: huge `PUSTR/PVSTR` (stress arrived broken from the turbulence scheme) vs. a collapsed `PTSKM1M`/large fluxes (the buoyancy amplifier). *This is the current open front for the mesh-increment crash family.*

1.20 Two more crash faces: a mid-year FESOM flux blow-up and a leg-start numerical explosion (2026-07-30)

The `voskin:282` front (§1.19) is OIFS-side and fires on step 1. Two *new* crash faces surfaced this week, both on the *FESOM* side, and neither is `voskin:282`.

(i) A mid-year FESOM flux blow-up. A colleague's 17-cycle concurrent run (`repro_movcav45`) died in FESOM's own `check_blowup`, not OIFS: cycle 19 (1918), `mstep` 14458 (mid-year, $\sim$day 201), at an *Arctic* node (lat $+83.5°$, lon $-53°$). `temp` hit 227°C, driven by a garbage surface heat flux `hflux`$= -4.0 \times 10^6\,\mathrm{W/m^2}$ ($\sim 10^4\times$ any physical air–sea flux) at an ice-edge node where sea ice vanished in a single step (`m_ice` $1.27 \to 0$). Only the surface layer blew up (every layer below normal, $-1.4 \ldots - 0.4$°C) and CFL stayed < 1 throughout, so this is a *surface-flux injection*, not an interior/velocity instability. The submesh the coupler built for that leg was *clean* — cavity front stable at $\sim 704\,\mathrm{km}$ every year (1910–13), matching PISM's $\sim 690\,\mathrm{km}$ — so the PISM→FESOM geometry conversion (`mesh_flags.py` + `fesom_submesh.x`) is *correct*; the bad flux is delivered by the atmosphere/OASIS, not made by a mesh error. Distinct from `voskin:282` (OIFS overflow, Antarctic, step 1); this is FESOM, Arctic, mid-year. The run's output was monthly / yearly-split with `NLOGPRT=1` and `EXPORTED` (not `EXPOUT`), so nothing flushed before the crash and the delivered flux at that step could not be recovered from disk — which set up the instrumented rerun.

The instrumented crash-hunt build (`fgdbg02`). To catch the flux *at ingest*: a `[FLUXGUARD]` added to `check_blowup` (`write_step_info.F90`) that, every step, prints `mstep, node, lon/lat, qso/qsi/qsr, a_ice/m_ice` for any node whose ingested `oce_flx_h`/`ice_flx_h` exceeds 5000 W/m^2 or is NaN; a *daily, daily-split* diagnostic stream of the atmosphere→ocean/ice flux components (`qsi/qso/qsr` plus the full `fh` decomposition `fh_sen/fh_lat/fh_radtot/fh_swr/fh_lwr/fh_lwrout` and `uwice/vwice`) with `split_freq=1d` so each day lands on disk before a mid-year blow-up (the previous config's monthly, yearly-split output is exactly why nothing survived); and FESOM `omp_num_threads=1` for bit-reproducibility, so a single crashing leg can later be re-run deterministically. Run for 100 yr because the blow-up timing is random.

(ii) A leg-start numerical explosion — a *third*, distinct face. `fgdbg02` blew up at cycle 7 (1906), but at `mstep` **1** — the very first timestep of the leg — at an *Antarctic ice-shelf* node (lat $-72.9°$, lon $-100°$, Amundsen Sea). `temp` went $2.8 \to -3.7\times10^{243}$ in one step (a full numerical explosion, top ∼17 levels, not a physical 227°C overshoot); `hflux`$= -90.5$ W/m^2 (*normal*), `hnode`/`Kv`/`W`/ `CFL_z` all normal, and `[FLUXGUARD]` stayed **silent**. The field that stood out in the dump is the SSH-solver right-hand-side: `ssh_rhs`$= 7.1\times10^6$ (from -208 the previous step), which is what this entry originally read as an *external-mode (SSH) equation* blow-up. ~~The couple-in hands FESOM a transport–geometry mismatch that detonates the SSH solve on step 1.~~ That reading did not survive opening the blow-up dump: `ssh_rhs`$\sim 10^6$ is the *ordinary* value at a cavity-front node and the real fault is elsewhere entirely (§1.21). What this entry does establish and what still stands: the run is on `which_ale='linfs'` (`step_per_day=72`, a 1200 s step), the scheme §1.15 showed *preserves* the cavity where `zstar` collapses it, so face (iii) is **not** the `zstar` seam collapse and `linfs` does not cure it; it is *not* advection, *not* `hnode` (§1.14, fixed; clean here), and *not* the ALE-mode issue of §1.15. The daily stream captured nothing (crash before day 1 closed) and the guard correctly stayed silent, validating the instrumentation while telling us this crash is not flux-driven. The `omp=1` build makes it bit-reproducible on one leg, which is what made §1.21 possible.

Three faces now, to keep separate. (i) `voskin:282`: OIFS single-precision overflow, chunk ≥ 2, Antarctic, step 1. (ii) mid-year FESOM flux blow-up: Arctic, ice-edge, a garbage OASIS-delivered heat flux. (iii) leg-start FESOM SSH-RHS explosion: Antarctic ice-shelf, *normal* flux, `mstep` 1, `ssh_rhs`$= 7.1\times10^6$ under `linfs`. Faces (i) and (iii) share step-1 timing and the Antarctic cavity region and may be the same underlying couple-in seeding; (ii) is a separate atmosphere-side flux corruption. The `omp=1` build makes face (iii) bit-reproducible on one leg — the cheapest next target.

1.21 Bug VIII — an unassigned array element in the GM streamfunction (2026-07-31)

Face (iii) is solved, and it is not an SSH problem. The 2 GB `fesom.1906.oce.blowup.nc` that `check_blowup` writes — a full 3D dump of `ssh_rhs`, `d_eta`, `u`, `v`, `u_rhs`, `helem`, `hnode`, `w`, `cfl_z`, `temp`, `salt` at the crash step — had never been opened. It settles the face in one pass.

The SSH framing dies first. `ssh_rhs`$= 7.1\times10^6$ is an unremarkable value here: 1461 nodes exceed 10^6, 506 exceed 10^7, the global max is 5.97×10^7 (8× the crash node), and the crash node ranks **143rd of the 184** pinched open nodes. `d_eta` stays within $[-1.76, +0.07]$ m everywhere — the CG solve absorbs the large rhs. There is no SSH-solver explosion. The large `ssh_rhs` at the front is itself structural: `eta` is identically 0 at all 2416 cavity nodes (FESOM's own convention, present in the native end-of-year restarts too) while the open ocean south of 60°S sits at -1.58 m, so each of the 1311 mixed cavity/open elements carries a ∼1.59 m `eta` step and a $-g\nabla\eta\,dt \approx 1.5$ m/s velocity increment. Median `max|u_rhs|` over those 1311 front elements is **1.01** m/s against 0.006–0.007 for open-ocean and full-cavity elements; the crash node's elements (1.11–1.25) sit between the p50 and p90 of that population. They are typical front elements.

The needle. Exactly **three** elements in 404 607 carry an interior exact `0.0` in `u` where `v` is non-zero (none in `v`, none in `helem`), and they touch exactly the five nodes whose tracers exploded. The same test over the campaign's thirteen saved restart states across seven cycles returns **zero**. A probe called at seven points of the timestep fired only *after* `solve_tracers_ale` — silent after `compute_vel_rhs`, `viscosity_filter`, `impl_vert_visc_ale`, `update_vel` on both sides of `exchange_elem`, `compute_hbar_ale` and `vert_vel_ale`.

Root cause. `fer_solve_Gamma` (`oce_fer_gm.F90`) solves the Gent–McWilliams streamfunction over `nzmin=ulevels_nod2D_max(n)` to `nzmax=nlevels_nod2D_min(n)` — max and min over the node's *surrounding elements*. A node touching both a deep-cavity element and a very shallow one inverts that range. At node 19343 (elements with `ulevels` $= [7, 1, 1, 1]$ and `nlevels` $= [18, 18, 14, 5]$) it is `nzmin`$= 7$, `nzmax`$= 5$. Both sweeps then run zero times, but

```
tr(:,nzmax) = tp(:,nzmax)
```

executes unconditionally and publishes an **unassigned element of the local automatic array** `tp` straight into `fer_gamma`. `fer_gamma2vel` divides gamma differences by `helem` to build the bolus velocity, giving the measured `fer_uv`$(1) = \mp 1.636 \times 10^{251}$ at levels 4 and 5 — equal magnitude, opposite sign, zonal component only, with `fer_uv`$(2) \sim 10^{-4}$ and `helem`$= 10\,\mathrm{m}$ both healthy, exactly the signature of one corrupted gamma at the level-5 interface. `solve_tracers_ale` then adds `fer_uv` to `UV`, **advects the tracers with it** (`temp`$\to 10^{250}$, `salt` pinned at its 3/45 bounds) and subtracts it back; $(x + y) - y$ with $|y| \gg |x|$ returns exactly `0.0`, which is the half-zeroed `u`. The identical add/subtract pair on `Wvel` with `fer_Wvel` five lines below produces the `W`$= 0$ holes. So the chain runs the *opposite* way to the first reading: the exact zeros are residue of the event, not its seed.

Why now, after a decade of FESOM. The trigger needs a node touching both a deep-cavity element and a very shallow one — a cavity front on steep bathymetry, which the moving cavity manufactures and relocates every cycle. The predicate max(`ulevels`) > min(`nlevels`) over a node's elements finds 1/2/1/2/0/2 such nodes in submeshes 1901–1906 and 2 in the base mesh, so the configuration recurs without always being fatal: `tp(:,nzmax)` is whatever the stack holds, usually benign, occasionally 10^{252}. That is the 1-in-hundreds behaviour of §1.20, and it is why the identical source crashed at `mstep` 1 under `-O3` but ran clean under `-check bounds -check pointers -check uninit -g`. `-check bounds` cannot catch it: `fer_gamma` is written in bounds; the defect is the *read of an unassigned value*. Requires `Fer_GM=.true.` (set here) and, for the degenerate range, `use_cavity=.true.`

Fix and verification. A node with no interior range carries no GM transport, so zero its column and skip:

```
if (nzmin >= nzmax) then; tr(:,:)  = 0.0_WP; cycle; end if
```

The leg had died at `mstep` 1 on every previous attempt, bit-identically; with the fix it ran to day 27 / step 1920 with both detectors silent and no blow-up. Upstream as FESOM/fesom2 PR #960 (branched off `main`, which carries the same defect), cherry-picked onto `awiesm3-implicit-ice-surftemp` as `26907ba9`.

TAINTED: cycles 1901–1905 ran with this active. Wherever a degenerate node existed and the stack word was small-but-non-zero, the GM bolus transport there was junk without crashing, so the GM contribution in those legs is unreliable even though they completed.

2 State of the cycle and of the climate (2026-07-27)

2.1 What is proven

- **Chunk-1 is quantitatively healthy and, under the coupled slab, crash-free**: January SST physical, zero translocated cells; OIFS Weddell September ci 0.61/0.99 (was 0.04); both-hemisphere sea-ice annual cycles; per-basin cavity melt in the observed regimes (§1.9); kilowatt flux events at real ice tiles: zero.

- **The workflow hands over in both directions** (Bug V fixed): awiesm3 → PISM → couple_out → chunk-3 couple_in runs unattended.

- **The PISM decade runs on FESOM's own melt** (`-ocean given`, 2026-07-17): real output, and a *gentle* geometry response — 290 fully-opened nodes vs 1399 in the Bug-I era, mean $|\Delta\text{draft}|$ 1.0 m, submesh 208 810 → 208 438 nodes (Fig. pism_change_mc33).

- **The live mesh change works**: weights regenerated and runtime-order-validated ($\leq 0.024°$), restarts remapped through the rpart sandwich, ICMGG/plit regenerated.

- **The increment test passes end-to-end** (2026-07-17/18): chunk-3 runs its full year on the changed mesh and hands to chunk-4 PISM unattended; movcav34 (10 awiesm3 years / 100 PISM years) repeats chunks 1–4 cleanly on the minimal committed build. The second increment's Ross-front blowup (§1.10) is the current front.

- **The high, rising cavity melt is diagnosed** (§1.12): near-realistic thermal forcing (+0.14 K) through a standard, validated melt scheme ($r = 0.91$), driven by cavity currents $\sim 6\times$ too fast (median 13 cm/s).

- **Those fast currents are injected by the mesh increment, not the physics** (§1.13–1.14, 2026-07-19/20): density-independent (chunk-1 / smoothed / ambient / *equilibrium* water all identical), geometry-independent (cold start on the changed mesh is healthy ∼2.6 cm/s). Tied to **the remapped `hnode`** (ALE layer thickness): `remap_hnode`'s ALE-rescale compresses the changed columns using the stale chunk-1 η. The `remap_hnode` edit (nominal at changed columns) is implemented and *halves* the over-speed (17 → 10 cm/s), but does not fully restore health — the residual is the **zstar/rigid-lid coordinate seam** (stretched open-ocean layers vs. nominal cavity layers). **`linfs` removes it cleanly**: the raw remap (no `hnode` surgery) under `linfs` does not spin up (∼1.7 cm/s). **Open:** a fresh coupled movcav run on `linfs` from chunk-1 to confirm end-to-end and check the open-ocean climate; strip the PGFDBG probe first.

- ~~The chunk-3 dilution enters through the vertical-advection top boundary (zero-tracer inflow ratchet / material-interface flux).~~ Cavity guards in both the implicit (`adv_tra_vert_impl`) and explicit (`upw1`, `qr4c`) schemes changed the crash bitwise-nothing; the guards were reverted. The identical-fraction $(T, S) \to (0, 0)$ signature belongs to the *multiplicative* `hnode` mismatch drain (Bug VI).

- ~~The dilution is unclosed column continuity: remnant $\nabla\cdot(u\,h)$ at pinned-η cavity columns drains tracer content at fixed volume.~~ The divergence and ice-base w are symptoms of the same `hnode` inconsistency, not an independent leak; with the restart `hnode` reset the columns are quiet.

- **The coupled-slab configuration is now correct on all `develop*` setups and the sea-ice climate validates** (§1.18, 2026-07-27): the OIFS `NAMMCC` switches (`LNEMOLIMTHK=.true.`, `LNEMOLIMTEMP=.false.`) are set in `awiesm3.yaml` (`0aca2487`); a 10-year `develop` run matches an independent CMIP7 spin-up hemisphere-by-hemisphere (NH March volume 27, obs ∼28–30), with residual biases NH-thin (∼20%) and SH-cycle-too-extreme. The `movcav` runs were never affected (their runscripts set `NAMMCC` by hand).

- **Open — the mesh-increment crash family** (`voskin:282`, Antarctic, chunk ≥ 2; §1.19): every restart/ICMGG input checked clean, so the overflow is generated in-step; OIFS instrumented, reproduction pending.
- **Open — the single-point thickness runaway**: a ∼52 m (by year 10) point in the `develop` sea-ice run, present *identically* in the reference CMIP7 spin-up (855 m by year 500); a shared FESOM point-convergence artifact, not coupling-specific.

2.2 Open item: the Southern-Hemisphere summer melt-back

Resolved in `movcav37` — see §1.17. The historical framing that identified the problem is kept below.

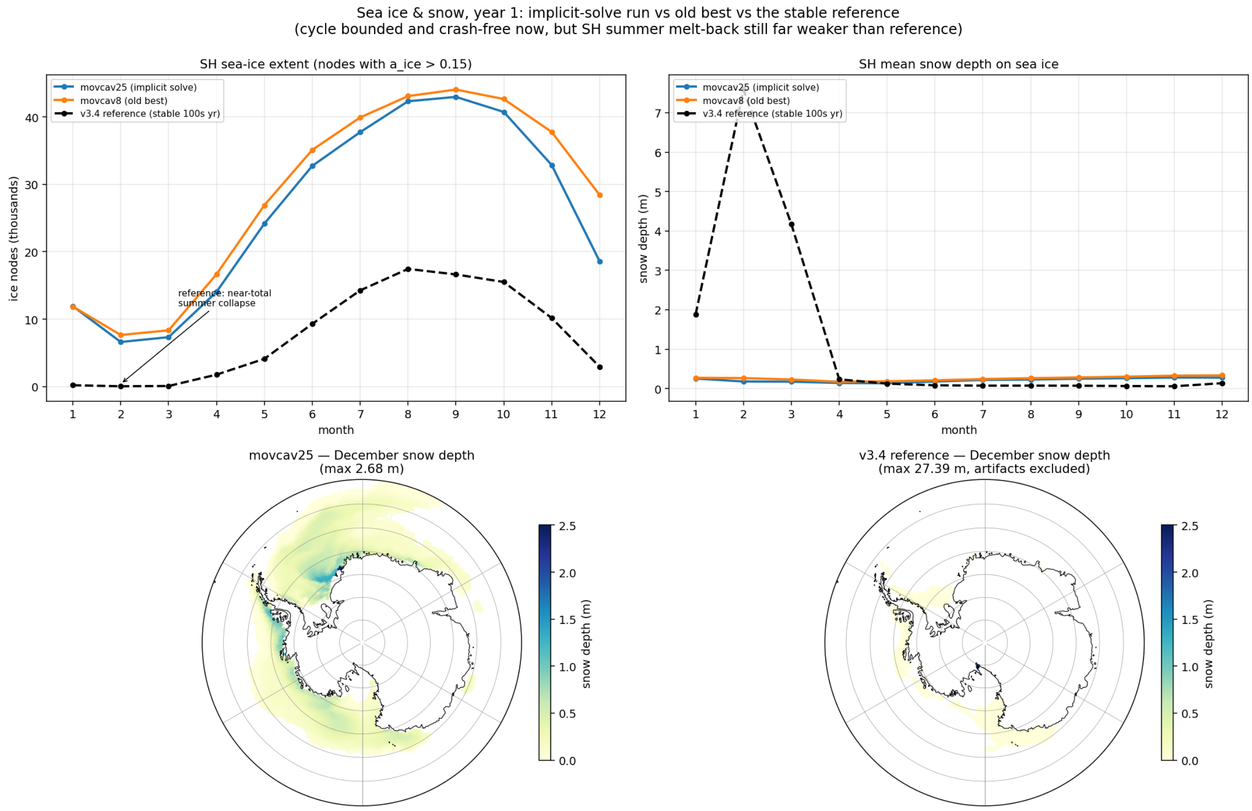

Figure 21: Year-1 sea ice and snow: implicit-solve run (blue) vs the movcav16-era state (orange) vs the 500-year-stable v3.4 reference (black dashed). Our runs track each other — the crash fixes did not degrade climate — but the reference collapses its SH ice almost completely each summer while ours retains ∼7k nodes, and its winter pack carries 0.03–0.08 m of snow against our 0.2–0.3 m. Bottom maps (same scale): December snow depth — our thick band hugs the Antarctic coast; the reference is nearly bare.

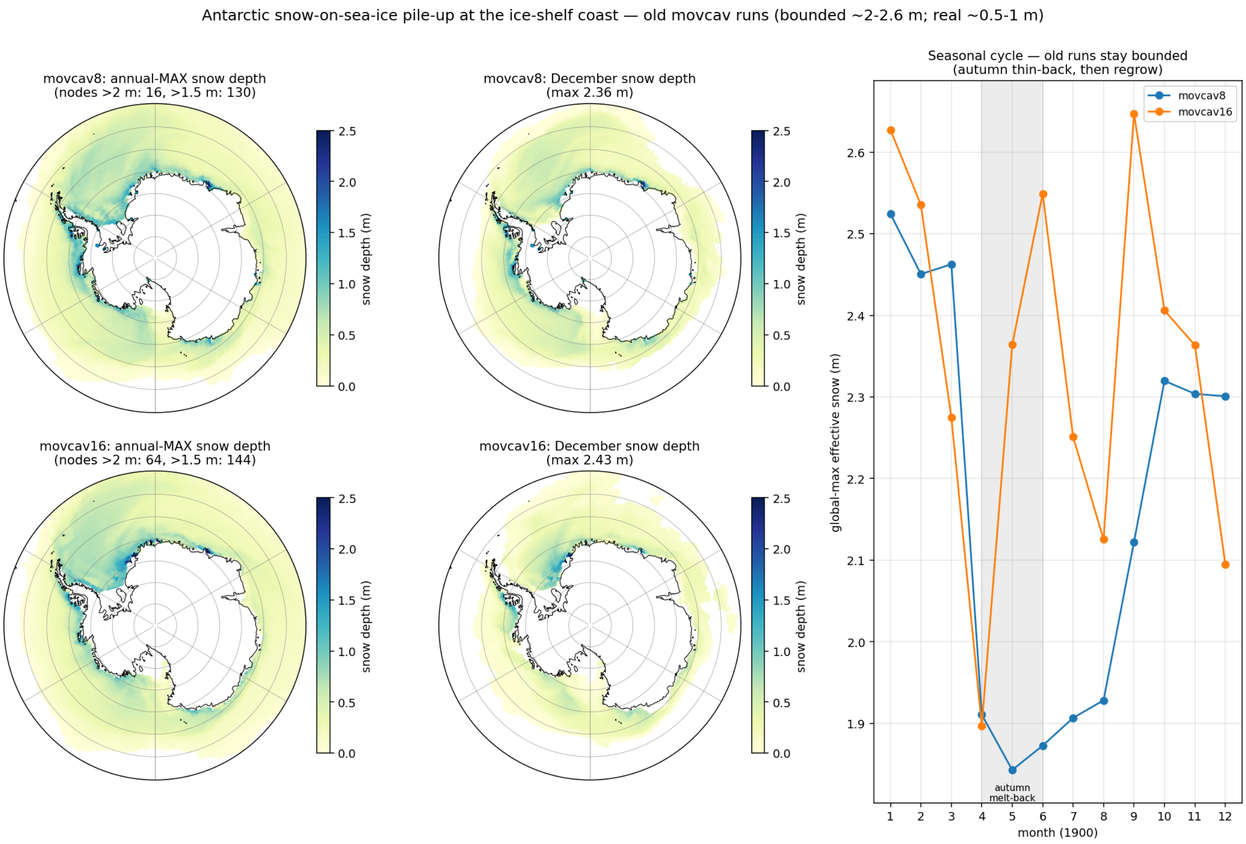

Figure 22: The extent of the coastal pile-up in the (pre-fix) movcav runs: annual-max and December snow depth, and the seasonal cycle. The thick snow is a band hugging the ice-shelf front (eastern Weddell/Fimbul, Amundsen), bounded at 2–2.6 m by an autumn melt-back. Later runs that lost this melt-back drifted monotonically to 3.5 m — the insulation that raised Bug IV's loop gain.

The ice-thermodynamics code is byte-identical to the v3.4 reference; the difference is state and forcing. Ours keeps ∼2.5× the reference's winter SH extent and only a ∼6× summer retreat where the reference manages ∼200× (Fig. 21), and piles 2–3.5 m of snow against the Antarctic ice-shelf coast (Fig. 22). Two mitigations exist: a *clean chunk-1 ice initial state* (reference year-1849 fields index-mapped onto the submesh via `map_nod.out`, 7 Ronne artifact nodes hybridized; snow max 0.81 m — in use since movcav29), and the coupled slab itself (the pinned-skin era's fictitious equilibria drove a spurious congelation-growth term). Whether these restore the reference's melt-back is the first climate question for the post-slab runs; meltponds were checked and are *not* the SH lever (the `hsno>hs1` lockout makes old and corrected schemes identical there).

3 Secondary bugs, fixed along the way

Bug	Fix	Commit
Remap extrapolated unboundedly below the old seabed (manufactured −45°C) when columns deepen	take water from the nearest old node wet at that depth	`eef4f6db`
Sub-ice η/`hbar` kept open-ocean values at newly-sub-ice nodes	zeroed (real bug, not the cure)	`b17fb5c8`

Bug	Fix	Commit
First mesh-change leg read the mother mesh as cavity-free (`cavity_nlvls.out` missing from the `mother_as_old` links) ⇒ zeros in newly-wet levels, mstep-1 blowup	link the cavity files	functions
Same-chunk `couple_in` rerun promoted its own submesh to `previous_submesh` (old = new ⇒ zeros again)	rerun guard	`b9c1e370`
OASIS `GSSPOS` conservation blowup on near-zero target residuals	`gauswgt_gsmart` transform	`3e94d4da`
namcouple `feom` dimension never actually patched (env var from another subjob's shell)	fixed variable passing	`3cf29e9e`
`fesom2ice` loaded the static full mesh against submesh output	load the run mesh	`63cfcc91`
`previous_submesh` read but never created	created; plus restart size guard	`3b44a704`
Failed coupling functions did not stop the workflow (backgrounded, observe saw no exit code)	`.coupling_failed` sentinel honoured by observe	`7953585b`
rstos size guard read the wrong dimension	fixed	`61c8ed8a`
ocp-tool flipped a coastline cell's mask+soil only, leaving an inconsistent surface column	masked nearest-neighbour rebuild of the full column	ocp-tool PR #51
Non-short-circuit `.and.` read `ulevels(0)` at every boundary edge (`ice_maEVP.F90`, both EVP variants); inside `if (use_cavity)`, so a garbage value decided whether sea-ice velocity was zeroed at the cavity–ocean edge	guard `edge_tri(2,ed)>0` separately	PR #960
Non-short-circuit `.and.` read `angles(0)` in the XIOS cell-corner insertion sort (`io_xios.F90`); harmless at `-O3`, aborts any bounds-checked build at init	split the loop test	PR #960
`alb` missing from `io_xios_is_ice_field` although declared `detect_missing_value="true"` ⇒ sea-ice albedo time-means averaged over open water	add `'alb'` to the ice-field list	PR #961

The recurring pattern: the guards were right and the plumbing around them defeated them — a correct refusal followed by an unconditional continue is worse than no guard at all.

4 Falsified claims (kept, struck through)

1. ~~Geostrophically-unbalanced remapped state; initialize changed cells in thermal-wind balance.~~ Failed its own offline gate (discrete ∇p noisier than NN fill); ships opt-in only, off by default.

2. ~~The sub-ice ssh BC violation is the blowup.~~ Real bug, fixed, run still died.

3. ~~The per-leg PISM increment is gentle ($\approx$22 m mean).~~ The mean was diluted by the 98% unchanged domain; the distribution is wildly skewed (max 923 m, 1399 fully-opened nodes).

4. ~~The salt-plume parameterization being off explains the warm cavity.~~ The spp-off cavity is the *coldest* in the CORE3 reference runs; the warm cavity was Bug II.

5. ~~The native ICMGG causes the mid-run crash family.~~ The step-0 swap tests proved the native, coastline-following surface set is *required* on the submesh (62 difference-zone cells cannot initialize otherwise); the mid-run family was Bug IV.

6. ~~Weighted-ist cures the crash family.~~ Necessary hygiene for the o2a blend, not sufficient: the pinned OIFS skin regenerated the instability from bounded inputs.

7. ~~The insulation/snow pile-up is the crash root.~~ It raises the loop gain (and is a real climate bias, §2), but the kW events occur on *thin* ice at 49°N; the clean-ini run crashed identically.

8. ~~Fragile $\leq$ 2-element coastal slivers cause the Ross-front blowup.~~ A control census killed it: the meshes that ran clean full years carry the same ~3450-node fragile count as the crashed one. The cause is increment magnitude, not coastline fragility (§1.11).

9. ~~The cavity melt gain is $\sim 10\times$ too steep (39 vs 1–4 m/yr/K).~~ The 1–4 figure is an integrated-shelf number, not the local exchange sensitivity; the standard three-equation scheme gives ~40 m/yr/K locally, so the measured 39 is correct. The melt formula is standard and reproduces the model's melt ($r = 0.91$); the excess is in the cavity *currents* (§1.12).

10. ~~The fast cavity currents are an under-damped equilibrium, cured by a tunable ice-base drag `C_d_cavity`.~~ At the stock drag, chunk-1 and *both* year-long cold-start equilibria run at the healthy ~2.6 cm/s; the fast currents are injected by the mesh increment, not a drag deficit (§1.13).

11. ~~The increment's geometry change injects the spin-up.~~ A cold start from rest on the changed mesh settles to ~2.6 cm/s — the new mesh is fine; the carried *remapped restart* is what is inconsistent with it (§1.13).

12. ~~The barotropic margin force F_x (eta+sea-ice) drives the spin-up.~~ The slow control carries the same F_x distribution (p90 $\sim 7 \times 10^{-4}$) as the fast runs; the step-1 force does not discriminate slow from fast (§1.13).

13. ~~The leg-start crash is an SSH-RHS explosion: the couple-in hands FESOM a transport–geometry mismatch that detonates the external-mode solve.~~ `ssh_rhs`$= 7.1 \times 10^6$ is ordinary at a cavity front — 506 nodes exceed 10^7, the crash node ranks 143rd of 184 pinched nodes, and `d_eta` stays ≤ 1.76 m everywhere because the CG solve absorbs it. The fault is in the GM streamfunction (§1.21).

14. ~~The pinched cavity-front peninsula tip drives the blow-up: remapped velocity has nowhere to go through the pinched faces.~~ The pinch is the normal front geometry and its `ssh_rhs` signature is unremarkable at the crash node. The pinch *is* implicated, but through `ulevels_nod2D_max` inverting the GM solve range, not through trapped transport (§1.21).

15. ~~The exact zeros in u and W seed the tracer explosion via a singular vertical operator.~~ `vert_vel_ale` runs *before* `solve_tracers_ale` and the probe after it is silent, so `W` was computed from a clean `UV`. Both zero fields are residue of the $(x + y) - y$ cancellation in the GM bolus add/subtract pair — the chain runs the other way (§1.21).

16. ~~The fix belongs in the restart remap (`mo_remap_fields.F90`).~~ The three affected elements come out of the remap clean and geometrically identical to the previous cycle (unchanged `ulevels`/`nlevels`, none new); `area`/`areasvol`, the partition and its halo comm arrays are all self-consistent. The defect is in FESOM's GM solver, exposed — not created — by the moving cavity (§1.21).

5 Method notes

1. **Cheap controls first.** The cold start (geometry vs state), running the fix, plotting the distribution instead of the mean — each killed a confident claim in minutes.

2. **Measurements from a corrupted run are worse than none** (Bug II poisoned Bug I's diagnosis for a week).

3. **"Same code as the stable reference" localizes nothing by itself** — the reference ran the same FESIM solve for 500 years; the difference was the state and, one layer deeper, a switch default that pinned a skin nobody remembered was pinned.

4. **Instrument both ends of a coupling before theorizing about either.** Receive-side logging (ICEDBG) proved the flux was corrupt; send-side decomposition (SENDDBG) named the term and the tile state in its first twelve lines. The two probes cost one rebuild each.

5. **Beware silent fallbacks** (Bug V's `isfile()` miss; the `-ocean given` silent fallback to `th`). A fallback on a nonexistent path should say so. The `NAMMCC` bug (§1.18) is the same shape: OIFS silently ran on default switches and discarded a coupled field for months, visible only in the sea-ice climate.

6. **`oasis3mct.lresume=false` silently destroys throughput.** With it false the coupling restart is rebuilt every exchange and the step rate decays ~100× over a run (0.28 → 11.4 s/step) — no single-leg or one-month test is long enough to see it. Use `lresume=true` from a restart set that carries the current coupled fields (the shared pool `rst*` predate `sit_feom` and cannot be used with `lresume=true` until regenerated).

7. **In a coupled run FESOM is not forcing-driven.** `namelist.forcing`'s JRA55-do paths are vestigial (standalone only); the ocean surface fluxes come from OIFS through OASIS, so the climate is set by the OIFS GHG year (`NCMIPFIXYR`: 1990 for `develop` here, 1850 for the a270270 reference), not by the forcing files.

6 Assets

- Send/receive flux instrumentation: `SENDDBG` (OIFS `A_Q_ice` decomposition + tile state), `ICEDBG` (FESIM received flux + state at cold nodes), `CPLDBG` (OIFS ingest bounds) — all log-only, strip before merging.

- `oasis_expout_o2a` runscript flag: every o2a exchange dumped to netcdf (~20× slowdown; crash-window probes only).

- Clean chunk-1 ice initial state (`prep/clean_ice_ini`): reference year-1849 ice fields index-mapped to the submesh; `map_nod.out` makes such remaps exact.

- A 2-minute offline reproducer (standalone FESOM on the PISM-cavity submesh) and a stable full-mesh control.
- `validate_feom_weights.py`: aborts any leg whose exchange weights are in the wrong node ordering — would have caught Bug II on day one.
- `verify_balance.py`: offline balance/stability gate for remapped restarts.
- Chunk-1 artifacts pooled at `input/fesom2/pism_cavity_ini/` (login-node launches).
- Face-(iii) probe set (`notes/ssh_rhs_probes/`): `blowup_uvrhs.py`, `eta_seam.py`, `front_census.py`, `uv_holes.py`, `uniqueness.py`, `inverted_range.py` (the max(`ulevels`) > min(`nlevels`) mesh predicate), `read_areas.py`, `partition_check.py`, `comm_symmetry.py`. Working notes in `notes/ssh_rhs_leg_sta`
- `fesom_meshdiag` (a standalone utility already in the FESOM tree, `BUILD_MESHDIAG=ON`) dumps `area`/`areasvol` per level without running the model — how §1.21 cleared the control-volume hypothesis.
- The crash-step blow-up dump `fesom.1906.oce.blowup.CRASH1_KEEP.nc`: a full 3D state at `mstep` 1. `check_blowup` writes one on every blow-up and it is routinely ignored; opening it is what solved face (iii).